4. Circuite electrice cu parametri distribuiti - continuare

(linii electrice/lungi/de transmisie)

SUPORT DIDACTIC PENTRU DISCIPLINA – "FIZICA III – ELECTROMAGNETISM"

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Cuprins

- 4.0. Preliminarii
- 4.1. Linii electrice - introducere
- 4.2. Modelul liniei. Parametri linieici.
- 4.3. Ecuatiile liniilor (telegrafistilor)
- 4.4. Linia fara pierderi in regim tranzitoriu. Aplicatie – reflexia unei unde; reflexii si refractii
- 4.5. Bilantul de puteri pe linie. Consecinte.
- 4.6. Linia electrica in regim armonic permanent. Parametrii secundari.
- 4.7. Ecuatiile liniilor exprimate in functie de marimile la capete. Aplicatie: impedanta de intrare a unei linii. Transformatoare de impedanta.
- 4.8. Aplicatii: modele de circuit, cu parametri distribuiti si concentrati, pentru microcomutatoare de RF.

4.6. Linia electrica in regim armonic permanent

- O linie electrica functioneaza in regim armonic permanent (r.a.p) daca tensiunea si curentul sunt in fiecare punct al liniei, functii armonice (sinusoidale) de timp, cu aceeasi frecventa.

$$u(z,t) = U(z)\sqrt{2}\sin\big(\omega t + \varphi_u(z)\big) \overset{C}{\rightleftarrows} \underline{U}(z) = U(z)e^{j\varphi_u(z)}$$

$$i(z,t) = I(z)\sqrt{2}\sin\big(\omega t + \varphi_i(z)\big) \overset{C}{\rightleftarrows} \underline{I}(z) = I(z)e^{j\varphi_i(z)}$$

Ecuațiile de ordinul I

$$\begin{cases} -\dfrac{\partial u(z,t)}{\partial z} = R_l\, i(z,t) + L_l \dfrac{\partial i(z,t)}{\partial t} \\ -\dfrac{\partial i(z,t)}{\partial z} = G_l\, u(z,t) + C_l \dfrac{\partial u(z,t)}{\partial t} \end{cases} \overset{C}{\rightleftarrows} \begin{cases} -\dfrac{\mathrm{d}\underline{U}(z)}{\mathrm{d}z} = (R_l + j\omega L_l)\,\underline{I}(z) \\ -\dfrac{\mathrm{d}\underline{I}(z)}{\mathrm{d}z} = (G_l + j\omega C_l)\,\underline{U}(z) \end{cases}$$

4.6. Linia electrica in regim armonic permanent

Ecuațiile de ordinul II

$$\begin{cases} \dfrac{\mathrm{d}^2\underline{U}(z)}{\mathrm{d}z^2} = (R_l + j\omega L_l)(G_l + j\omega C_l)\underline{U}(z) \\ \dfrac{\mathrm{d}^2\underline{I}(z)}{\mathrm{d}z^2} = (R_l + j\omega L_l)(G_l + j\omega C_l)\underline{I}(z) \end{cases}$$

$\underline{\gamma}$ → constanta de propagare complexa a liniei

$$\underline{\gamma} \overset{\text{def}}{=} \sqrt{(R_l + j\omega L_l)(G_l + j\omega C_l)}$$

$$\underline{\gamma} = \alpha + j\beta$$

$\alpha = \Re(\underline{\gamma}) > 0$ → constanta de atenuare [Np/m]

$\beta = \Im(\underline{\gamma}) > 0$ → constanta de fază (nr. de unda) [rad/m]

α si β se pot calcula in functie de parametrii lineici:

$$\alpha = \sqrt{\frac{1}{2}\left[R_l\,G_l - \omega^2 L_l C_l + \sqrt{(R_l\,G_l - \omega^2 L_l C_l)^2 + \omega^2(L_l G_l + R_l C_l)^2}\right]}$$

$$\beta = \sqrt{\frac{1}{2}\left[-R_l\,G_l + \omega^2 L_l C_l + \sqrt{(R_l\,G_l - \omega^2 L_l C_l)^2 + \omega^2(L_l G_l + R_l C_l)^2}\right]}$$

α si β depinde de frecventa (prin ω explicit dar si prin parametrii lineici)

4.6. Linia electrica in regim armonic permanent

Rezolvam ecuatia de ordinul II in $\underline{U}(z)$v

$$\frac{\mathrm{d}^2\underline{U}(z)}{\mathrm{d}z^2} = \underline{\gamma}^2\underline{U}(z)$$

$$\underline{x}^2 - \underline{\gamma}^2 = 0 \Rightarrow \underline{x}_{1,2} = \pm\underline{\gamma}$$

$$\underline{U}(z) = \underline{C}_1 e^{-\underline{\gamma}z} + \underline{C}_2 e^{+\underline{\gamma}z}$$

unda elementara directa de tensiune = $\underline{U}_d(z)$

unda elementara inversa de tensiune = $\underline{U}_i(z)$

$$\underline{U}(z) = \underline{U}_d(z) + \underline{U}_i(z)$$

Curentul se determina dintr-o relatie de ordinul I

$$-\frac{\mathrm{d}\underline{U}(z)}{\mathrm{d}z} = (R_l + j\omega L_l)\,\underline{I}(z)$$

$$\underline{I}(z) = -\frac{1}{R_l + j\omega L_l}\frac{\mathrm{d}\underline{U}(z)}{\mathrm{d}z} = -\frac{1}{R_l + j\omega L_l}\left(-\underline{\gamma}\underline{C}_1 e^{-\underline{\gamma}z} + \underline{\gamma}\underline{C}_2 e^{+\underline{\gamma}z}\right)$$

$$\underline{I}(z) = \frac{\underline{\gamma}}{R_l + j\omega L_l}\left(\underline{C}_1 e^{-\underline{\gamma}z} - \underline{C}_2 e^{+\underline{\gamma}z}\right)$$

$$\underline{I}(z) = \frac{1}{\underline{Z}_c}\left(\underline{U}_d(z) - \underline{U}_i(z)\right)$$

$$\underline{Z}_c \overset{\text{not}}{=} \frac{R_l + j\omega L_l}{\underline{\gamma}} = \frac{R_l + j\omega L_l}{\sqrt{(R_l + j\omega L_l)(G_l + j\omega C_l)}}$$

$$\underline{Z}_c = \sqrt{\frac{R_l + j\omega L_l}{G_l + j\omega C_l}}$$

Impedanța caracteristică complexă a liniei

4.6. Linia electrica in regim armonic permanent

Concluzii:

1)Solutia ecuatiilor liniei **cu** pierderi in regim armonic permanent:

$$\underline{U}(z) = \underline{U}_d(z) + \underline{U}_i(z)$$

$$\underline{I}(z) = \frac{1}{\underline{Z}_c}\left(\underline{U}_d(z) - \underline{U}_i(z)\right)$$

2)Parametrii secundari ai liniei sunt

$\underline{Z}_c = \sqrt{\frac{R_l + j\omega L_l}{G_l + j\omega C_l}}$ - impedanta caracteristica complexa a liniei

$\underline{\gamma} = \sqrt{(R_l + j\omega L_l)(G_l + j\omega C_l)}$ - constanta de propagare complexa a liniei

Parametrii secundari depind de frecventa (explicit dar si implicit, prin parametrii lineici).

3) $\underline{U}_d(z) = \underline{C}_1 e^{-\underline{\gamma} z}$, $\underline{U}_i(z) = \underline{C}_2 e^{+\underline{\gamma} z}$ → unde elementare de tensiune;

$\underline{I}_d(z) = \frac{\underline{U}_d(z)}{\underline{Z}_c}$, $\underline{I}_i(z) = -\frac{\underline{U}_i(z)}{\underline{Z}_c}$ → unde elementare de curent

4) Constantele $\underline{C}_1$ si $\underline{C}_2$ se determina din conditiile concrete in care functioneaza linia.

5) *Reprezentarea in timp a undelor elementare =>*

4.6. Linia electrica in regim armonic permanent

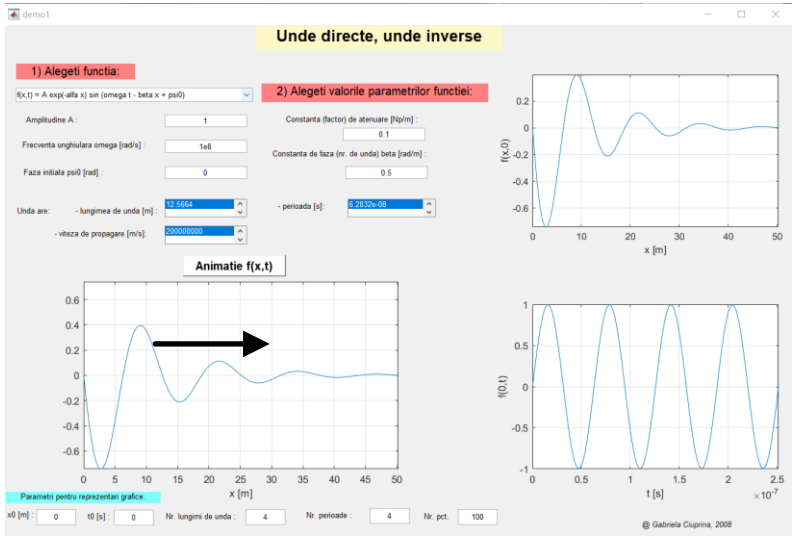
$$\underline{U}_d(z) = \underline{C}_1 e^{-\underline{\gamma} z} \qquad \underline{U}_d(0) = \underline{C}_1 \triangleq U_{d0}\, e^{j\,\varphi_0}$$

$$\underline{U}_d(z) = U_{d0} e^{j\,\varphi_{d0}} e^{-(\alpha + j\beta)z} = U_{d0} e^{-\alpha z} e^{j\,(-\beta z + \varphi_{d0})} \quad \rightleftarrows \quad u_d(z,t) = U_{d0} e^{-\alpha z} \sqrt{2} \sin(\omega t - \beta z + \varphi_{d0})$$

faza

$\omega T = 2\pi$ $\qquad u_d(z,t) = u_d(z, t+T)$

$\beta\lambda = 2\pi$ $\qquad u_d(z,t) e^{-\alpha\lambda} = u_d(z+\lambda, t)$



Vedeti demo matlab – demo1.m

Punctele de faza constanta:

$$\omega t - \beta z + \varphi_{d0} = \text{constant}$$

$$\omega\, \mathrm{d}t - \beta\, \mathrm{d}z = 0$$

$$\frac{dz}{dt} = \frac{\omega}{\beta} = v$$

Viteza de propagare

$$v = \frac{\omega}{\beta} = \frac{2\pi}{T}\frac{\lambda}{2\pi} \Rightarrow v = \frac{\lambda}{T} \Rightarrow \qquad \lambda = vT$$

4.6. Linia electrica in regim armonic permanent

Parametrii secundari in cazuri particulare:

a) Linia fara pierderi $(R_l = 0; G_l = 0)$

$$\underline{\gamma} = \sqrt{(R_l + j\omega L_l)(G_l + j\omega C_l)} = \sqrt{-\omega^2 L_l C_l} = j\,\omega\sqrt{L_l C_l}$$

$$\underline{\gamma} = \alpha + j\beta$$

$\alpha = 0$ Nu exista atenuare

$$\beta = \omega\sqrt{L_l C_l}$$

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{L_l C_l}}$$

Daca parametrii lineici nu depind de frecventa, atunci viteza de propagare nu depinde de frecventa. => Semnalele se propaga fara distorsiuni (toate armonicele se propaga cu aceeasi frecventa).

$$\underline{Z}_c = \sqrt{\frac{L_l}{C_l}} \in \mathrm{R}\,;\ \underline{Z}_c \overset{\text{not}}{=} Z_0$$

Impedanta caracterstica este un numar real.

b) Linia cu pierderi dar fara distorsiuni

Conditii pentru propagare fara distorsiuni:
1. α sa nu depinda de ω
2. v sa nu depinda de ω

$$\underline{\gamma} = \sqrt{(R_l + j\omega L_l)(G_l + j\omega C_l)} = \sqrt{L_l C_l\left(\frac{R_l}{L_l} + j\omega\right)\left(\frac{G_l}{C_l} + j\omega\right)}$$

$$\underline{\gamma} = \left(\frac{1}{\tau} + j\omega\right)\sqrt{L_l C_l}$$

in conditia Heaviside $\boxed{\frac{R_l}{L_l} = \frac{G_l}{C_l} \overset{\text{not}}{=} \frac{1}{\tau}}$

Conditia Heaviside poate fi indeplinita prin pupinizare (inserierea deperiodica de bobine pe linie)

https://en.wikipedia.org/wiki/Loading_coil

$$\begin{cases} \alpha = \frac{1}{\tau}\sqrt{L_l C_l};\ \frac{1}{\tau^2} = \frac{R_l G_l}{L_l C_l} \Rightarrow \alpha = \sqrt{R_l G_l} \\ \beta = \omega\sqrt{L_l C_l} \Rightarrow v = \frac{\omega}{\beta} = \frac{1}{\sqrt{L_l C_l}} \end{cases}$$

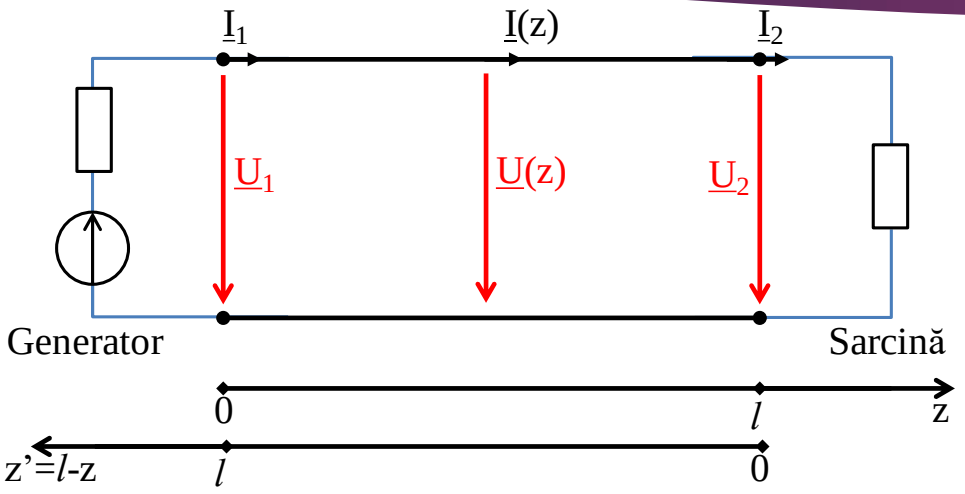
Exista atenuare, dar ea nu depende de ω

Viteza de propagare nu depende de ω

$$\underline{Z}_c = \sqrt{\frac{R_l + j\,\omega\,L_l}{G_l + j\,\omega\,C_l}} = \sqrt{\frac{\frac{R_l}{L_l} + j\,\omega}{\frac{G_l}{C_l} + j\,\omega}\,\frac{L_l}{C_l}} = \sqrt{\frac{L_l}{C_l}} \in \mathrm{R}\,;\ \underline{Z}_c \overset{\text{not}}{=} Z_0$$

Impedanta caracterstica este un numar real.

4.7. Ecuatiile liniilor exprimate in functie de marimile la capete (r.a.p)



Notatii

$$\underline{U}(0)=\underline{U}_1 \quad \underline{U}(l)=\underline{U}_2$$
$$\underline{I}(0)=\underline{I}_1 \quad \underline{I}(l)=\underline{I}_2$$

$$\begin{cases} \underline{U}(z)=\underline{C}_1 e^{-\underline{\gamma}z}+\underline{C}_2 e^{+\underline{\gamma}z} \\ \underline{I}(z)=\dfrac{1}{\underline{Z}_c}\left(\underline{C}_1 e^{-\underline{\gamma}z}-\underline{C}_2 e^{+\underline{\gamma}z}\right) \end{cases}$$

util

$\underline{\gamma}=\alpha+j\beta \Rightarrow \underline{\gamma}=j\beta;\ \underline{Z}_c \overset{not}{=} Z_0$

$\cosh(\underline{\gamma}z)=\cosh(j\beta z)=\cos(\beta z)$

$\sinh(\underline{\gamma}z)=\sinh(j\beta z)=j\cdot\sin(\beta z)$

Ecuatiile liniei in functie de marimile de intrare.

(Ideea: constantele $\underline{C}_1$si $\underline{C}_2$ trebuie exprimate in functie de $\underline{U}_1$si $\underline{I}_1$.)

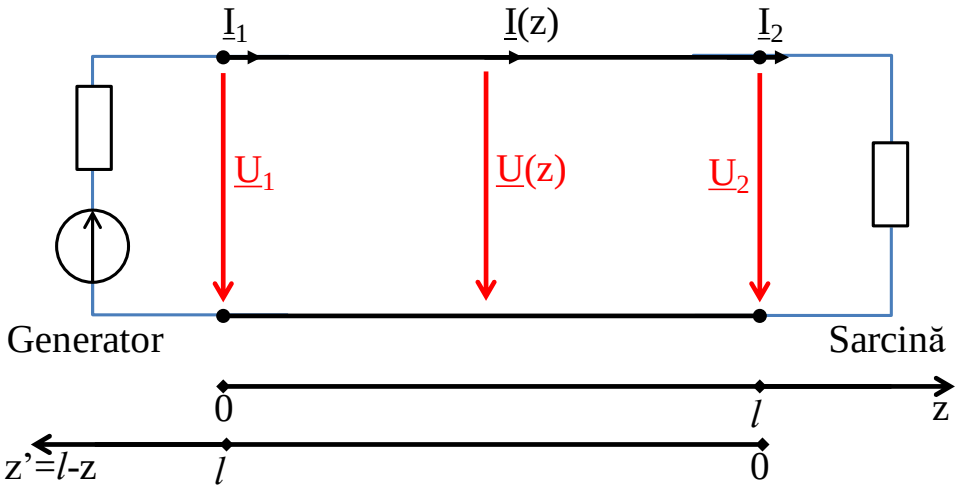
$$z=0 \Rightarrow \begin{cases} \underline{U}_1=\underline{C}_1+\underline{C}_2 \\ \underline{Z}_c\underline{I}_1=\underline{C}_1-\underline{C}_2 \end{cases} \overset{(+)}{\Rightarrow} \underline{C}_1=\frac{\underline{U}_1+\underline{Z}_c\underline{I}_1}{2};\ \overset{(-)}{\Rightarrow} \underline{C}_2=\frac{\underline{U}_1-\underline{Z}_c\underline{I}_1}{2}$$

$$\begin{cases} \underline{U}(z)=\underline{U}_1\cosh(\underline{\gamma}z)-\underline{Z}_c\underline{I}_1\sinh(\underline{\gamma}z) \\ \underline{I}(z)=\underline{I}_1\cosh(\underline{\gamma}z)-\dfrac{\underline{U}_1}{\underline{Z}_c}\sinh(\underline{\gamma}z) \end{cases}$$

$$\begin{cases} \underline{U}(z)=\underline{U}_1\cos(\beta z)-j\,Z_0\underline{I}_1\sin(\beta z) \\ \underline{I}(z)=\underline{I}_1\cos(\beta z)-j\,\dfrac{\underline{U}_1}{Z_0}\sin(\beta z) \end{cases}$$

Pentru o linie fara pierderi

4.7. Ecuatiile liniilor exprimate in functie de marimile la capete (r.a.p)



Notatii

$$\underline{U}(0)=\underline{U}_1 \quad \underline{U}(l)=\underline{U}_2$$
$$\underline{I}(0)=\underline{I}_1 \quad \underline{I}(l)=\underline{I}_2$$

$$\begin{cases} \underline{U}(z)=\underline{C}_1 e^{-\underline{\gamma} z}+\underline{C}_2 e^{+\underline{\gamma} z} \\ \underline{I}(z)=\dfrac{1}{\underline{Z}_c}\left(\underline{C}_1 e^{-\underline{\gamma} z}-\underline{C}_2 e^{+\underline{\gamma} z}\right) \end{cases}$$

Ecuatiile liniei in functie de marimile de iesire.

(Ideea: constantele $\underline{C}_1$ si $\underline{C}_2$ trebuie exprimate in functie de $\underline{U}_2$ si $\underline{I}_2$.)

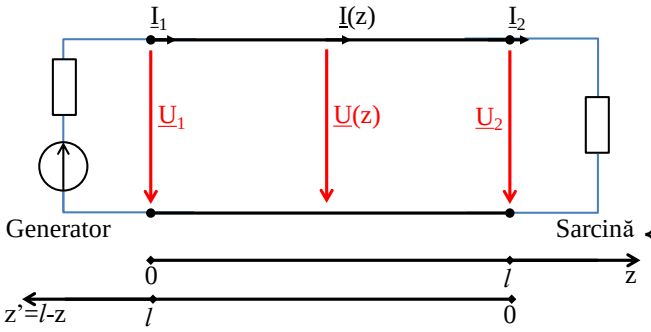
$$z=l \Rightarrow \begin{cases} \underline{U}_2=\underline{C}_1 e^{-\underline{\gamma} l}+\underline{C}_2 e^{+\underline{\gamma} l} \\ \underline{Z}_c\underline{I}_2=\underline{C}_1 e^{-\underline{\gamma} l}+\underline{C}_2 e^{+\underline{\gamma} l} \end{cases} \quad \underline{C}_1=\frac{\underline{U}_2+\underline{Z}_c\underline{I}_2}{2}e^{+\underline{\gamma} l} \quad \underline{C}_2=\frac{\underline{U}_2-\underline{Z}_c\underline{I}_2}{2}e^{-\underline{\gamma} l}$$

$$\begin{cases} \underline{U}(z')=\underline{U}_2\cosh(\underline{\gamma} z')+\underline{Z}_c\underline{I}_2\sinh(\underline{\gamma} z') \\ \underline{I}(z')=\underline{I}_2\cosh(\underline{\gamma} z')+\dfrac{\underline{U}_2}{\underline{Z}_c}\sinh(\underline{\gamma} z') \end{cases}$$

$$\begin{cases} \underline{U}(z')=\underline{U}_2\cos(\beta z')+j\,Z_0\underline{I}_2\sin(\beta z') \\ \underline{I}(z')=\underline{I}_2\cos(\beta z')+j\,\dfrac{\underline{U}_2}{Z_0}\sin(\beta z') \end{cases}$$

Pentru o linie fara pierderi

4.7. Ecuatiile liniilor exprimate in functie de marimile la capete (r.a.p)



Ecuatiile cuadripolare ale liniei.

* In functie de marimile de intrare

$$\begin{cases} \underline{U}(z) = \underline{U}_1 \cosh(\underline{\gamma} z) - \underline{Z}_c \underline{I}_1 \sinh(\underline{\gamma} z) \\ \underline{I}(z) = \underline{I}_1 \cosh(\underline{\gamma} z) - \dfrac{\underline{U}_1}{\underline{Z}_c} \sinh(\underline{\gamma} z) \end{cases}$$

* In functie de marimile de iesire

$$\begin{cases} \underline{U}(z') = \underline{U}_2 \cosh(\underline{\gamma} z') + \underline{Z}_c \underline{I}_2 \sinh(\underline{\gamma} z') \\ \underline{I}(z') = \underline{I}_2 \cosh(\underline{\gamma} z') + \dfrac{\underline{U}_2}{\underline{Z}_c} \sinh(\underline{\gamma} z') \end{cases}$$

$z = l$

$$\begin{cases} \underline{U}_2 = \underline{U}_1 \cosh(\underline{\gamma} l) - \underline{Z}_c \underline{I}_1 \sinh(\underline{\gamma} l) \\ \underline{I}_2 = \underline{I}_1 \cosh(\underline{\gamma} l) - \dfrac{\underline{U}_1}{\underline{Z}_c} \sinh(\underline{\gamma} l) \end{cases}$$

$z' = l$

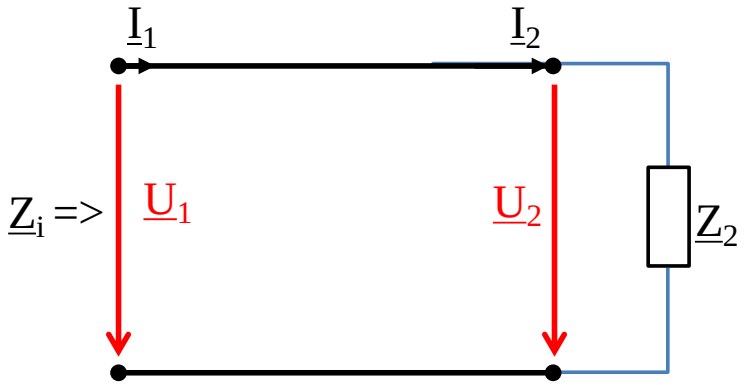
$$\begin{cases} \underline{U}_1 = \underline{U}_2 \cosh(\underline{\gamma} l) + \underline{Z}_c \underline{I}_2 \sinh(\underline{\gamma} l) \\ \underline{I}_1 = \underline{I}_2 \cosh(\underline{\gamma} l) + \dfrac{\underline{U}_2}{\underline{Z}_c} \sinh(\underline{\gamma} l) \end{cases}$$

$$\begin{cases} \underline{U}_2 = \underline{U}_1 \cos(\beta l) - j\, Z_0 \underline{I}_1 \sin(\beta l) \\ \underline{I}_2 = \underline{I}_1 \cos(\beta l) - j\, \dfrac{\underline{U}_1}{Z_0} \sin(\beta l) \end{cases}$$

Pentru o linie fara pierderi

$$\begin{cases} \underline{U}_1 = \underline{U}_2 \cos(\beta l) + j\, Z_0 \underline{I}_2 \sin(\beta l) \\ \underline{I}_1 = \underline{I}_2 \cos(\beta l) + j\, \dfrac{\underline{U}_2}{Z_0} \sin(\beta l) \end{cases}$$

4.7. Aplicatie – impedanta de intrare a unei linii



$$\underline{Z}_i \overset{\text{def}}{=} \frac{\underline{U}_1}{\underline{I}_1}$$

Folosim ecuatiiile cuadripolare ale liniei, in functie de marimile de iesire. Consieram linia fara pierderi.

$$\begin{cases} \underline{U}_1 = \underline{U}_2 \cos(\beta l) + j\, Z_0 \underline{I}_2 \sin(\beta l) \\ \underline{I}_1 = \underline{I}_2 \cos(\beta l) + j\, \dfrac{\underline{U}_2}{Z_0} \sin(\beta l) \end{cases}$$

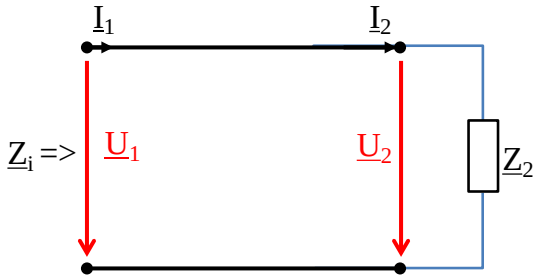
$$\underline{U}_2 = \underline{Z}_2 \cdot \underline{I}_2$$

$$\beta = \frac{2\pi}{\lambda}$$

$$\underline{Z}_i = \frac{\underline{Z}_2 \underline{I}_2 \cos(\beta l) + j Z_0 \underline{I}_2 \sin(\beta l)}{\underline{I}_2 \cos(\beta l) + \mathrm{j} \dfrac{\underline{Z}_2\, \underline{I}_2}{Z_0} \sin(\beta l)}$$

$$\underline{Z}_i = Z_0 \frac{\underline{Z}_2 \cos\left(\frac{2\pi l}{\lambda}\right) + j\, Z_0 \sin\left(\frac{2\pi l}{\lambda}\right)}{Z_0 \cos\left(\frac{2\pi l}{\lambda}\right) + j\, \underline{Z}_2 \sin\left(\frac{2\pi l}{\lambda}\right)}$$

4.7. Aplicatie – impedanta de intrare a unei linii



$$\underline{Z}_i = Z_0 \frac{\underline{Z}_2 \cos\left(\frac{2\pi l}{\lambda}\right) + j\, Z_0 \sin\left(\frac{2\pi l}{\lambda}\right)}{Z_0 \cos\left(\frac{2\pi l}{\lambda}\right) + j\, \underline{Z}_2 \sin\left(\frac{2\pi l}{\lambda}\right)}$$

$$\underline{Z}_i = Z_0 \frac{\underline{Z}_2 + j\, Z_0 \operatorname{tg}\left(\frac{2\pi l}{\lambda}\right)}{Z_0 + j\, \underline{Z}_2 \operatorname{tg}\left(\frac{2\pi l}{\lambda}\right)} = Z_0 \frac{\underline{Z}_2 \operatorname{ctg}\left(\frac{2\pi l}{\lambda}\right) + j\, Z_0}{Z_0 \operatorname{ctg}\left(\frac{2\pi l}{\lambda}\right) + j\, \underline{Z}_2}$$

$\underline{Z}_i$ depinde de $Z_0; \frac{l}{\lambda}; \underline{Z}_2$

I)dependența de λ pentru $\underline{Z}_2$ fixat

a) Scurt circuit $\underline{Z}_2 = 0$

$$\Rightarrow \underline{Z}_i = j\, Z_0 \operatorname{tg}\left(\frac{2\pi l}{\lambda}\right) \triangleq \underline{Z}_{i,o}$$

b) Gol $\underline{Z}_2 \to \infty$

$$\Rightarrow \underline{Z}_i = -jZ_0 \operatorname{ctg}\left(\frac{2\pi l}{\lambda}\right) \triangleq \underline{Z}_{i,\infty}$$

c) Linie adaptată $\underline{Z}_2 = Z_0$

$$\Rightarrow \underline{Z}_i = Z_0$$

Se spune ca linia adaptata este "transparenta", indiferent de lungimea ei.

Important pentru proiectarea circuitelor de microunde:

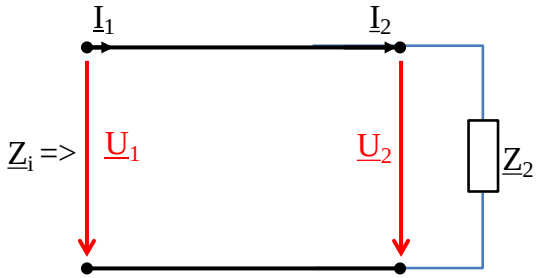
Liniile terminate in scurt sau gol se comporta la intrare, in functie de lungimea lor ca un element L sau C ideal, de orice valoare vrem, iar cele adaptate ca un rezistor cu $R = Z_0$

Important pentru metoda de masura a lui Z_0

$$\underline{Z}_{i,o}\underline{Z}_{i,\infty} = Z_0^2 \Rightarrow Z_0 = \sqrt{\underline{Z}_{i,o}\underline{Z}_{i,\infty}}$$

Impedanta caracteristica este media geometrica a impedantelor de intrare, masurate cu iesirea in gol si in scurt.

4.7. Aplicatie – impedanta de intrare a unei linii



$$\underline{Z}_i = Z_0 \frac{\underline{Z}_2 + j\, Z_0 \operatorname{tg}\left(\frac{2\pi l}{\lambda}\right)}{Z_0 + j\, \underline{Z}_2 \operatorname{tg}\left(\frac{2\pi l}{\lambda}\right)} = Z_0 \frac{\underline{Z}_2 \operatorname{ctg}\left(\frac{2\pi l}{\lambda}\right) + j\, Z_0}{Z_0 \operatorname{ctg}\left(\frac{2\pi l}{\lambda}\right) + j\, \underline{Z}_2}$$

$\underline{Z}_i$ depinde de $Z_0; \frac{l}{\lambda}; \underline{Z}_2$

a) **Linie de lungime multiplu întreg de semilungimea de undă**

$$l = k\frac{\lambda}{2} \Rightarrow \frac{2\pi l}{\lambda} = \frac{2\pi}{\lambda} \cdot k\frac{\lambda}{l} = k\pi \Rightarrow \operatorname{tg}(k\pi) = 0 \Rightarrow \underline{Z}_i = \underline{Z}_2$$

→ Linia se comporta ca un scurt-circuit între intrare și ieșire

II)dependența de $\underline{Z}_2$ pentru λ fixat

b) **Linie de lungime multiplu impar de sfert de lungimea de undă**

$$l = (2k+1)\frac{\lambda}{4} \Rightarrow \frac{2\pi l}{\lambda} = \frac{2k+1}{2}\pi \Rightarrow \operatorname{ctg}\left(\frac{2k+1}{2}\pi\right) = 0 \Rightarrow \boxed{\underline{Z}_i = \frac{Z_0^2}{\underline{Z}_2}}$$

→ Scurt la ieșire $\underline{Z}_2 = 0 \Rightarrow \underline{Z}_i \to \infty$; se comportă ca un gol

→ Gol la ieșire $\underline{Z}_2 = \infty \Rightarrow \underline{Z}_i = 0$; se comportă ca un scurt (foarte periculos pentru linii)

→ Bobină la ieșire $\underline{Z}_2 = j\omega L \Rightarrow \underline{Z}_i = \frac{Z_0^2}{j\omega L} = -j\frac{Z_0^2}{\omega L}$; comportare capacitiva

→ Condensator la ieșire $\underline{Z}_2 = \frac{1}{j\omega c} \Rightarrow \underline{Z}_i = j\omega C Z_0^2$; comportare inductiva

4.7. Aplicatie – impedanta de intrare a unei linii

Recomandare bibliografica:

https://www.rfcafe.com/references/qst/review-transmission-lines-circuit-elements-november-1966-qst.htm

A Review of Transmission Lines as Circuit Elements
November 1966 QST Article

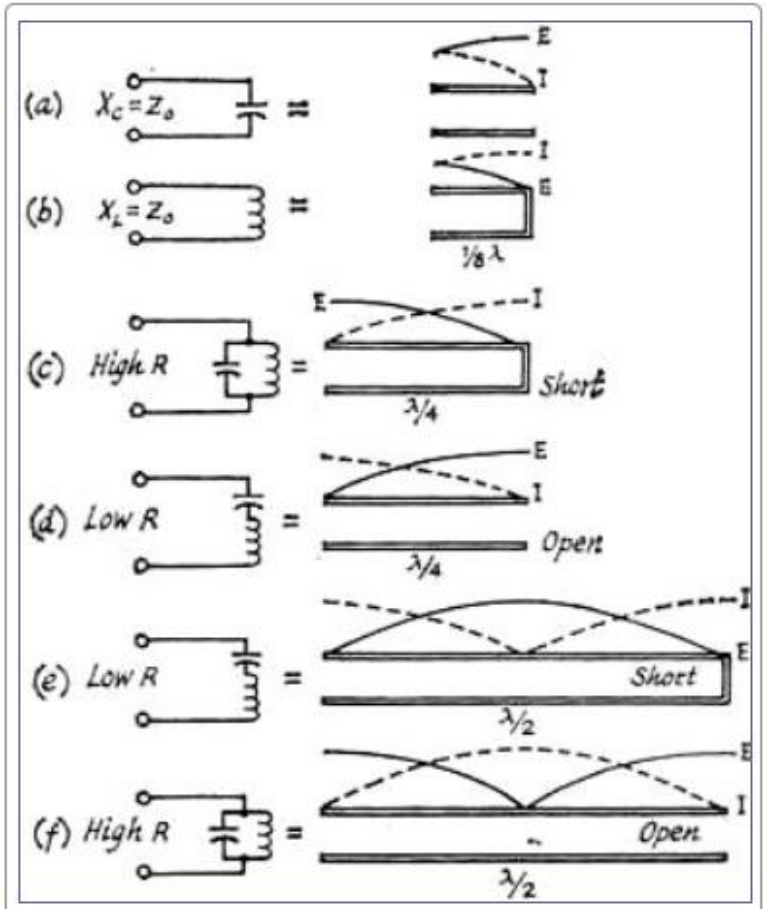


Fig. 1 - Table of equivalent circuits using resonant lines. Voltage and current relationships are illustrated for open and shorted line.

4.7. Aplicatie – transformatoare de impedanta

Rationamentele anterioare – utile pentru realizarea transformatoarelor de impedanta.

Ideea: Z_0

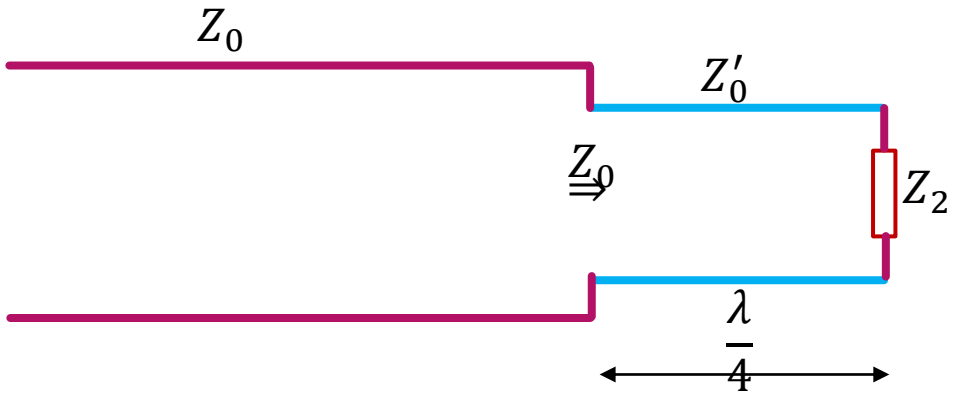


$Z_2 \neq Z_0$

Se folosesc pentru adaptarea la cuplarea a doua componente, de exemplu curcuitul RF (transmisie/receptie) la antena. Astfel creste eficienta transmisiei radio.

In acest caz apar reflexii ale semnalelor, pentru ca sarcina nu este adaptata liniei.

Solutie: se intercaleaza o linie de lungime $\frac{\lambda}{4}$, care este o sectiune de adaptare. Parametrii ei se aleg astfel incat la intrarea ei impedanta de intrare sa fie Z_0



$$Z_0 = \frac{(Z_0')^2}{Z_2} \quad \Rightarrow \quad Z_0' = \sqrt{Z_0\, Z_2}$$

Adaptarea este valabila numai la o frecventa data.
Se spune ca traf.de impedanta este un dispozitiv de banda ingusta, selectiv.

https://www.emtalk.com/mwt_mpa.htm

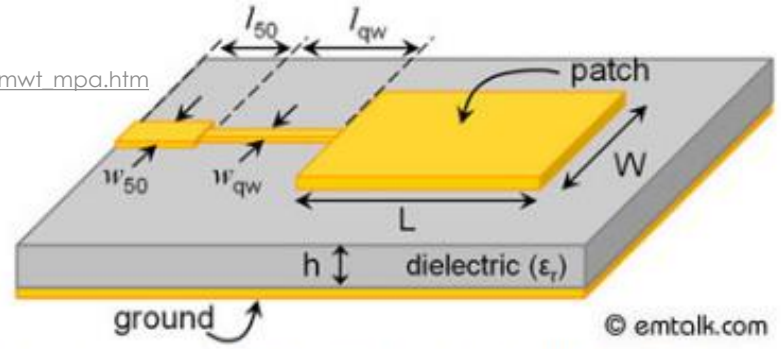
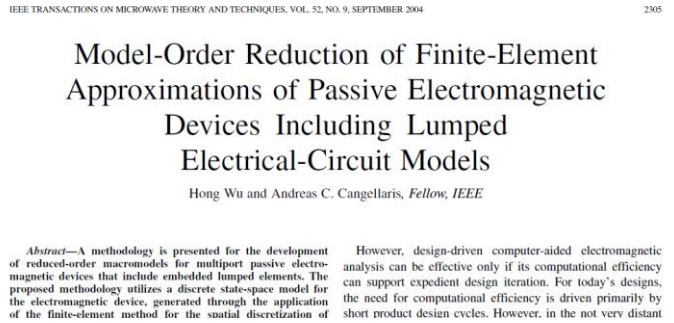


Figure 1. Model of microstrip patch antenna; edge-fed with quarter wavelength transformer section to 50 Ω transmission line.

4.7. Aplicatie – calculul impedantei de intrare – studiu de caz

Inspirat din articolul:

Hong Wu and A. C. Cangellaris, "Model-order reduction of finite-element approximations of passive electromagnetic devices including lumped electrical-circuit models," in *IEEE Transactions on Microwave Theory and Techniques*, vol. 52, no. 9, pp. 2305-2313, Sept. 2004, doi: 10.1109/TMTT.2004.834582.



IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES, VOL. 52, NO. 9, SEPTEMBER 2004 2305

Model-Order Reduction of Finite-Element Approximations of Passive Electromagnetic Devices Including Lumped Electrical-Circuit Models

Hong Wu and Andreas C. Cangellaris, *Fellow, IEEE*

Abstract—A methodology is presented for the development of reduced-order macromodels for multiport passive electromagnetic devices that include embedded lumped elements. The proposed methodology utilizes a discrete state-space model for the electromagnetic device, generated through the application of the finite-element method for the spatial discretization of

However, design-driven computer-aided electromagnetic analysis can be effective only if its computational efficiency can support expedient design iteration. For today's designs, the need for computational efficiency is driven primarily by short product design cycles. However, in the not very distant

The first example considered is a **terminated air-filled coaxial cable**. The **length of the coaxial cable is 1 m** and its two electrodes are assumed to be perfectly conducting. **The radius of the inner circular cylindrical electrode is 4 mm. The inner radius of the outer circular cylindrical electrode is 8 mm**. The cable is terminated at its **far end by a lumped circuit that includes a 100-pF capacitor in series with the parallel combination of a 5- resistor with a 10-nH inductor**. In addition, on the **driving end of the cable, a 5- shunt resistor is present connecting the two electrodes.** A TL model for this system is straightforward to set up and analyze. For the purposes of macromodeling, what is of interest **is the frequency-dependent input impedance of the resulting circuit.**

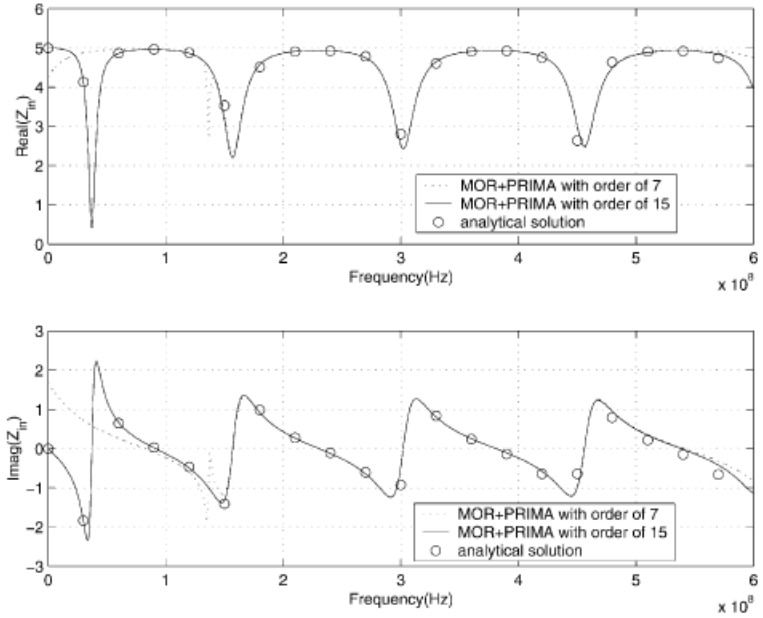


Fig. 1. Input impedance of a terminated coaxial cable.

4.7. Aplicatie – calculul impedantei de intrare – studiu de caz

Conductoarele sunt perfecte => $\sigma_c \to \infty$ => $R \to 0$, cablul este fara pierderi.

$l = 1\ \text{m}$
$a = 4\ \text{mm}$
$b = 8\ \text{mm}$
$\sigma_c \to \infty$
$\sigma_d = 0$
$\varepsilon_d = \varepsilon_0$
$\mu_d = \mu_0$

Parametrii geometrici si de material

Parametrii lineici, dedusi din rationamente de regim stationar. (Vedeti si P4)

$$R_{l,cc} = \frac{1}{\sigma\pi}\left(\frac{1}{a^2} + \frac{1}{c^2 - b^2}\right) \to 0$$

$$L_{l,ext} = \frac{\mu_0}{2\pi}\ln\frac{b}{a} \qquad L_{l,int} = \frac{\mu_0}{8\pi}$$

$$C_l = \frac{2\,\pi\varepsilon_d}{\ln\frac{b}{a}} \qquad G_l = \frac{2\,\pi\sigma_d}{\ln\frac{b}{a}} = 0$$

Parametrii lineici

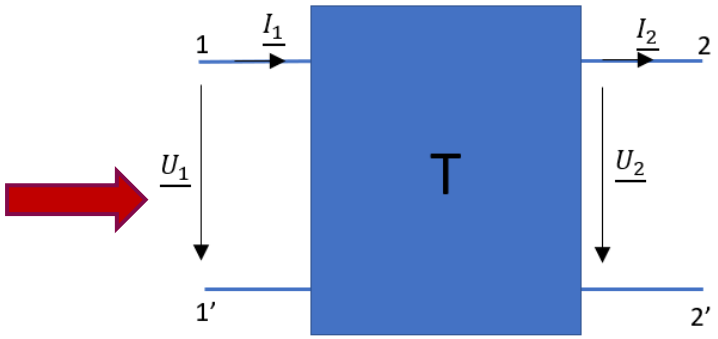
$$\underline{\gamma} = \sqrt{(R_l + j\omega L_l)(G_l + j\omega C_l)}$$

$$\underline{Z}_c = \sqrt{\frac{R_l + j\omega L_l}{G_l + j\omega C_l}}$$

Parametrii secundari

4.7. Aplicatie – calculul impedantei de intrare – studiu de caz

Ecuatiile cuadripolare ale liniei, in functie de marimile de iesire (aici relatiile generale).



$$\begin{cases} \underline{U}_1 = \underline{U}_2 \cosh(\underline{\gamma} l) + \underline{Z}_c \underline{I}_2 \sinh(\underline{\gamma} l) \\ \underline{I}_1 = \underline{I}_2 \cosh(\underline{\gamma} l) + \dfrac{\underline{U}_2}{\underline{Z}_c} \sinh(\underline{\gamma} l) \end{cases}$$

$$\begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \underline{U}_2 \\ \underline{I}_2 \end{bmatrix}$$

Matricea de transfer

$$A = \cosh(\underline{\gamma} l)$$

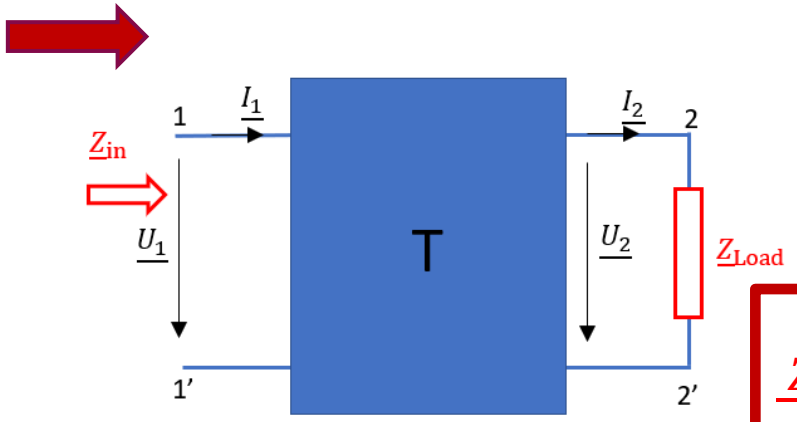
$$B = \underline{Z}_c \sinh(\underline{\gamma} l)$$

$$C = \frac{1}{\underline{Z}_c} \sinh(\underline{\gamma} l)$$

$$D = \cosh(\underline{\gamma} l)$$

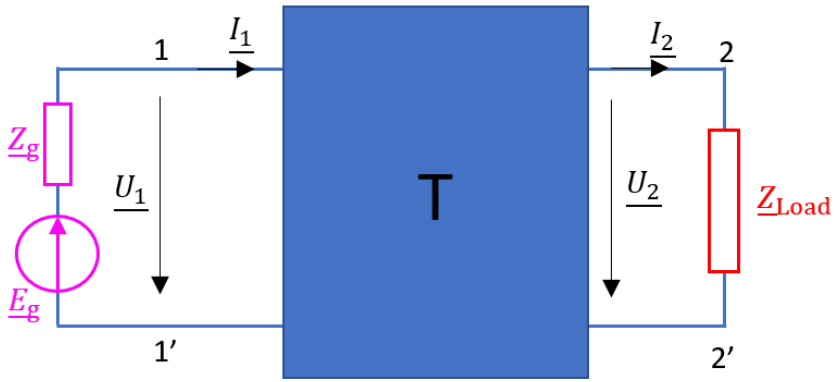
Conexiunea cu restul circuitului

4.7. Aplicatie – calculul impedantei de intrare – studiu de caz

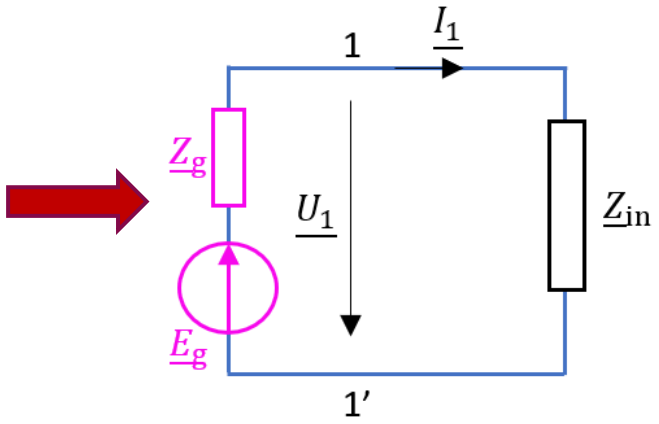


$$\begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} U_2 \\ I_2 \end{bmatrix}$$

$$U_2 = Z_{\text{Load}}\, I_2$$



$$Z_{\text{in}} = \frac{U_1}{I_1} = \frac{A\, U_2 + B\, I_2}{C\, U_2 + D\, I_2} = \frac{A\, Z_{\text{Load}} + B}{C\, Z_{\text{Load}} + D}$$



marimile de intrare

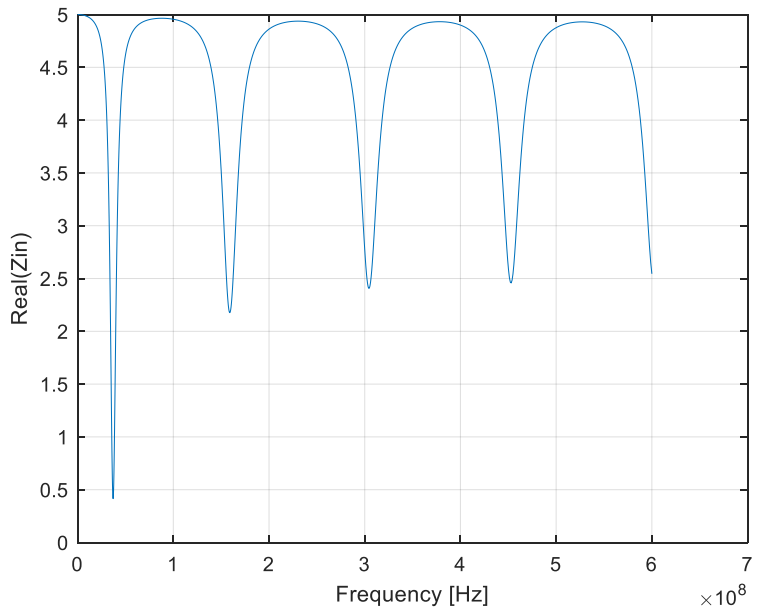
$$I_1 = \frac{E_g}{Z_g + Z_{\text{in}}}$$

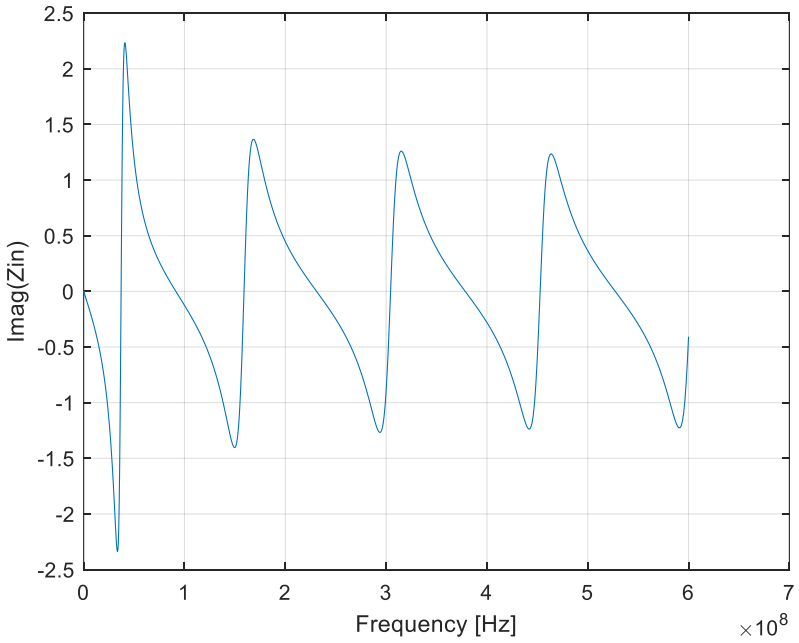
$$U_1 = Z_{\text{in}}\, I_1$$

marimile de iesire

$$\begin{bmatrix} U_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} \begin{bmatrix} U_1 \\ I_1 \end{bmatrix}$$

4.7. Aplicatie – calculul impedantei de intrare – studiu de caz



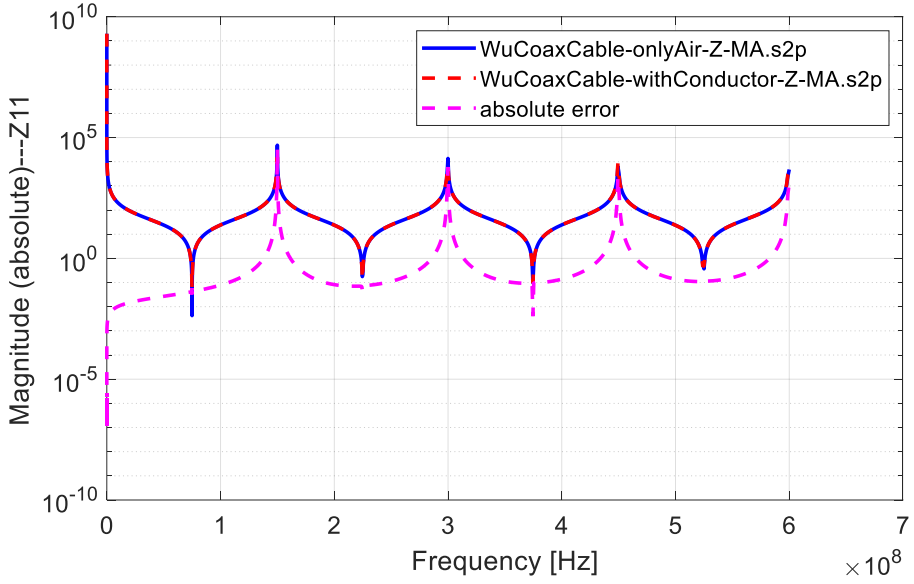


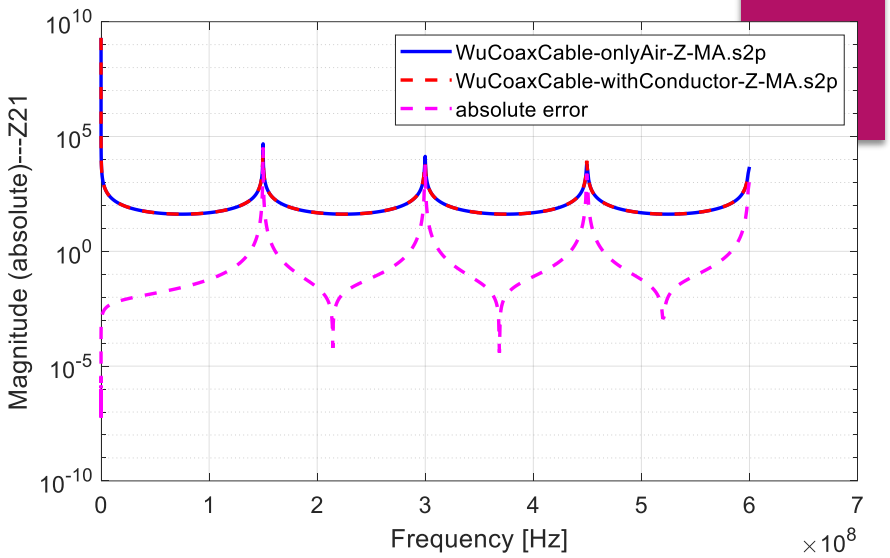
Cod Matlab disponibil.
Include inclusiv o varianta cu piederi in conductoarele liniei, si parametri lineici dependent de frecventa.

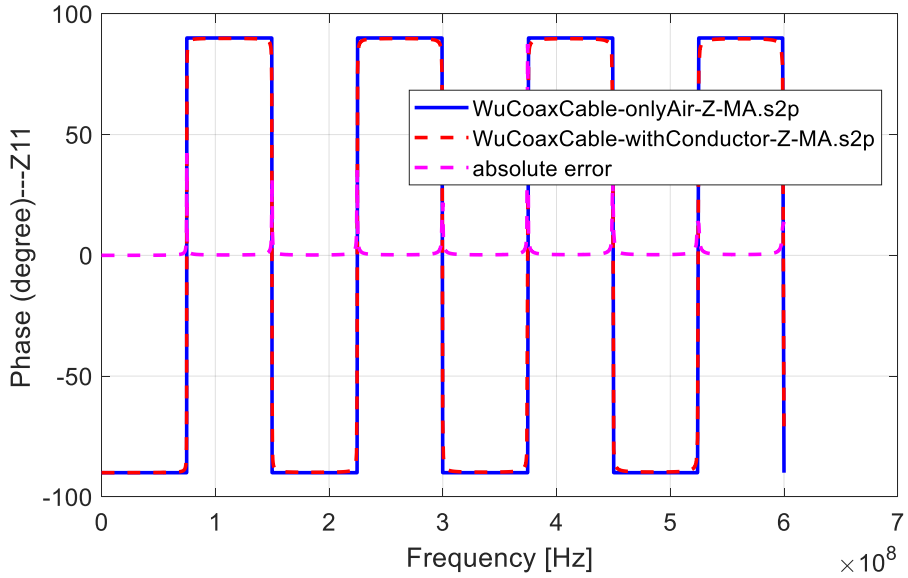
Se salveaza si matricile Z si Y (2 x 2), in format Touchstone.

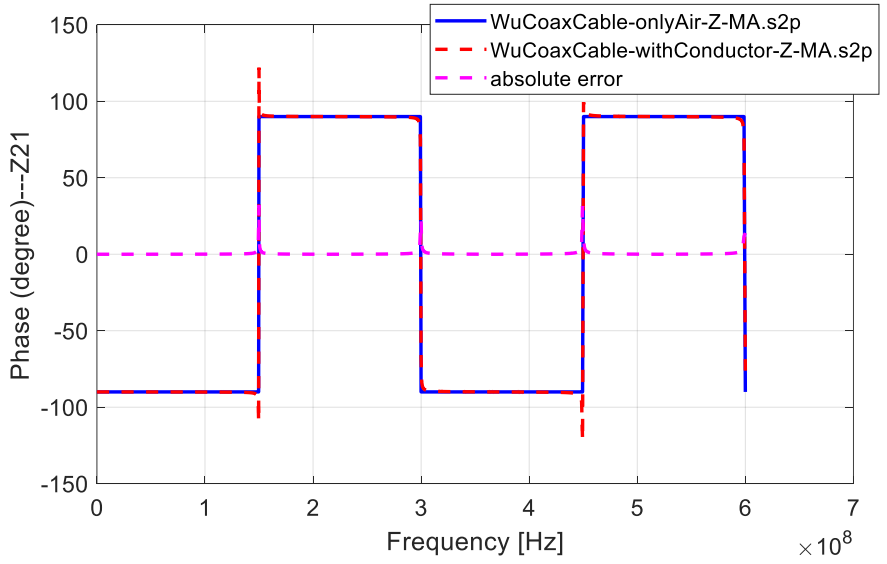
Pentru inspectare – folositi snpdiff_tool in linia de comanda.

Parametrii Z ai liniei – modul si argument, linia fp si cazul adaugarii pierderilor in conductor

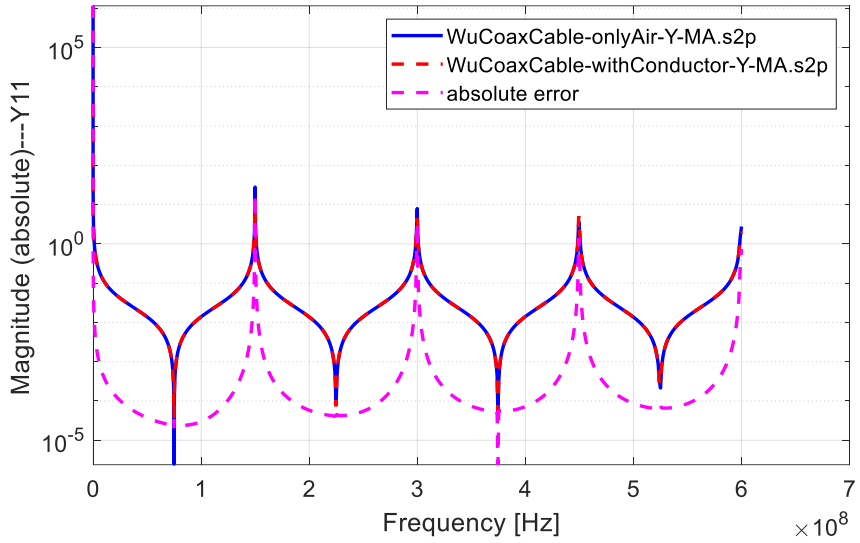


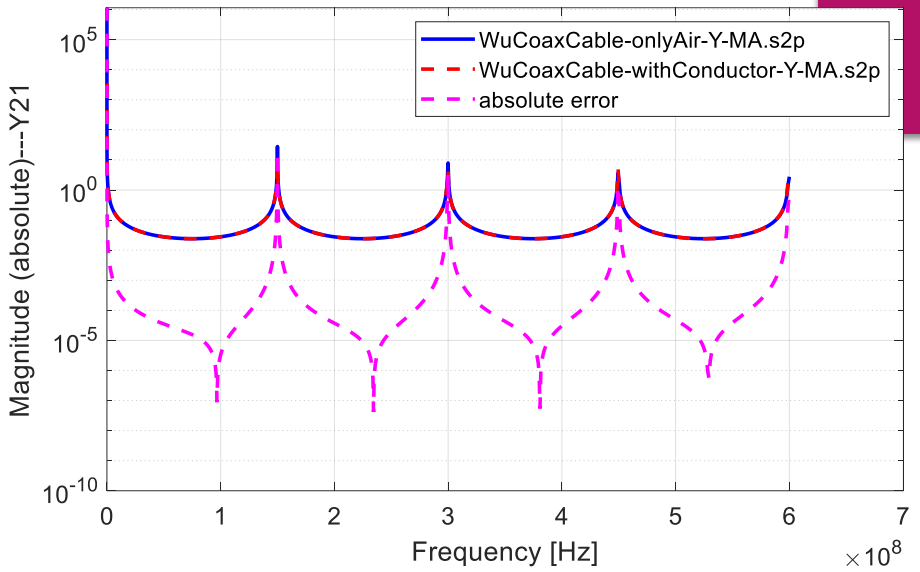


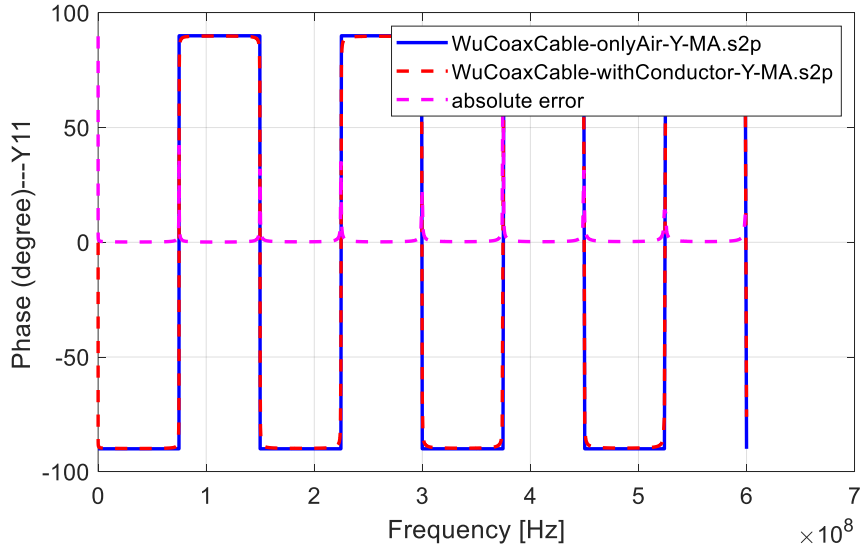


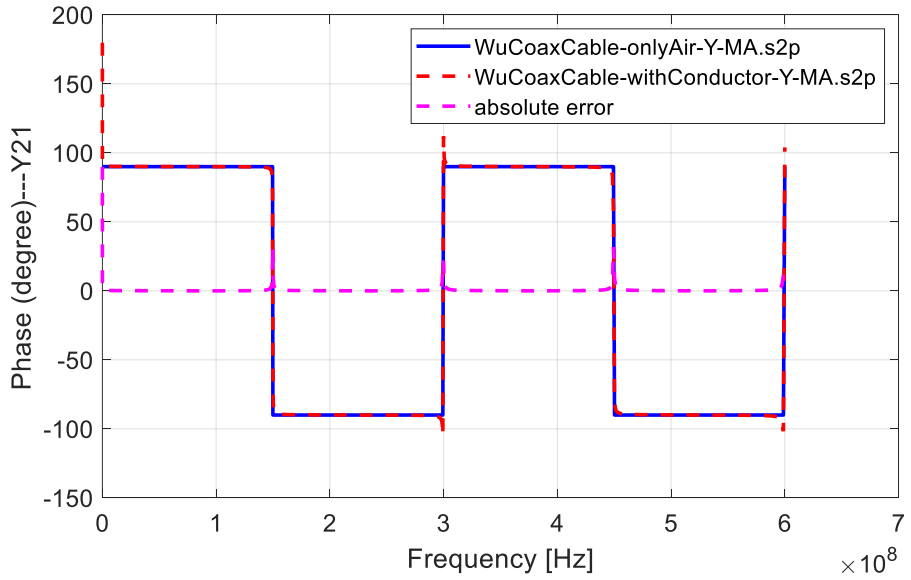


Parametrii Y ai liniei – modul si argument, linia fp si cazul adaugarii pierderilor in conductor

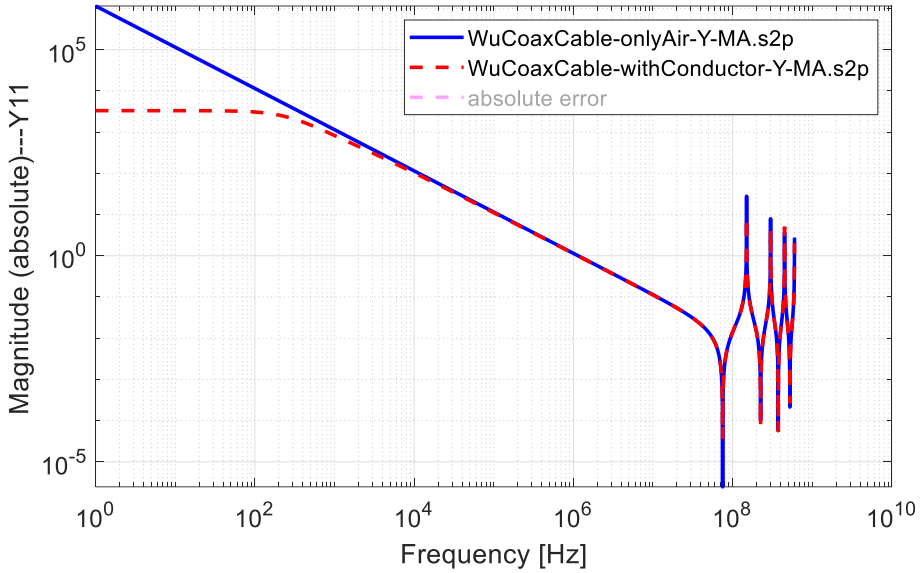


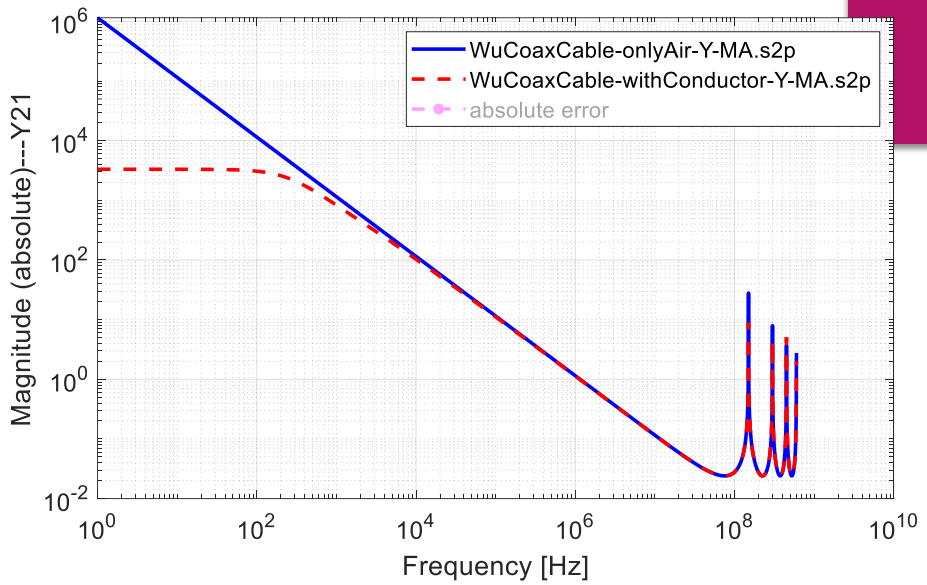


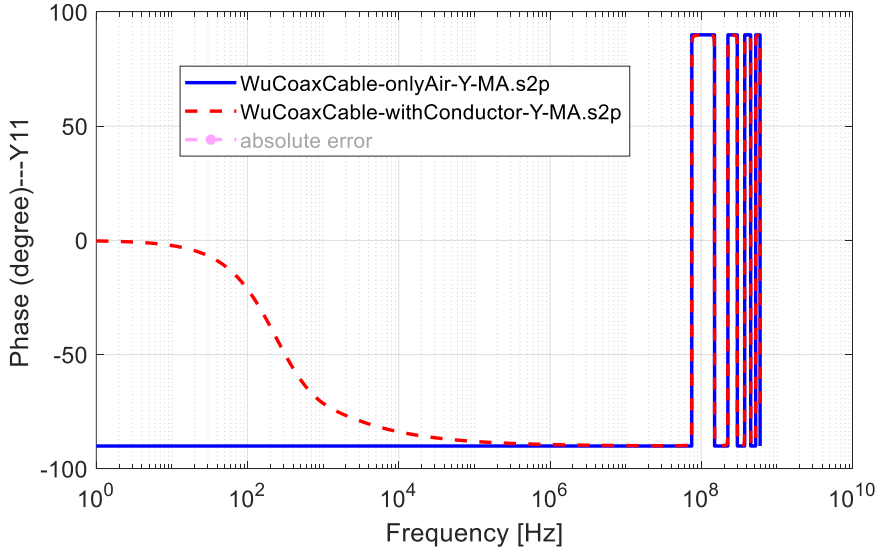


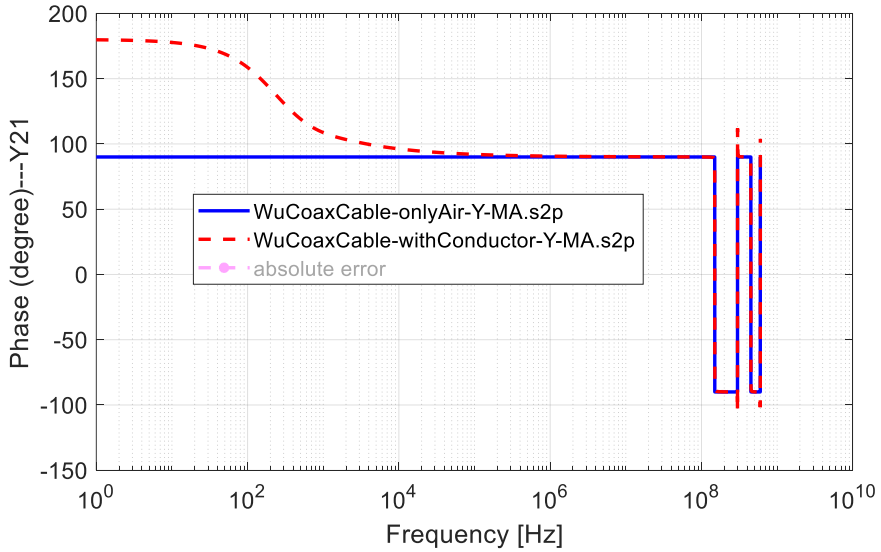


Y – ca in slide-ul anterior, dar scara logaritmica pentru frecvente









Cod Matlab disponibil.
Poate fi folosit pentru a intelege legatura dintre lungimea liniei, frecventa maxima si numarul de rezonante.

4.7. Aplicatii – linii de transmisie in circuite cu microunde

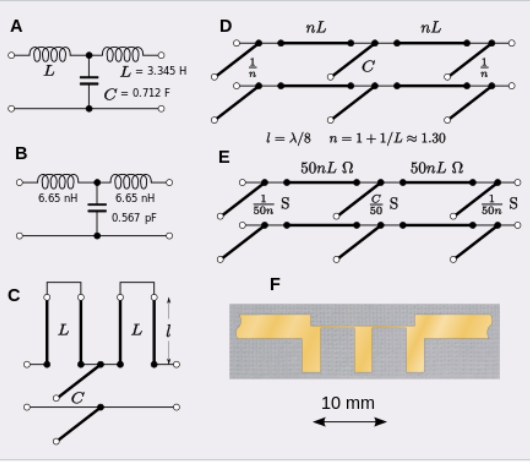
David S. Ricketts – An overview of transmission line circuit elements

https://rickettslab.org/bits2waves/design/wilkinson-2/introduction-to-transmission-line-circuits/

Applications: Filters, dividers/combiners, couplers

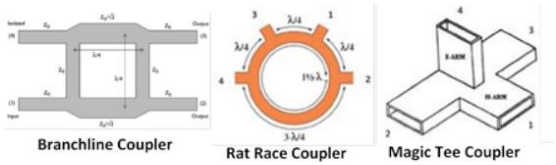
Commensurate line circuit

https://en.wikipedia.org/wiki/Commensurate_line_circuit



are electrical circuits composed of transmission lines that are all the same length; commonly one-eighth of a wavelength. Lumped element circuits can be directly converted to distributed-element circuits of this form by the use of Richards' transformation. This transformation has a particularly simple result; inductors are replaced with transmission lines terminated in short-circuits and capacitors are replaced with lines terminated in open-circuits. Commensurate line theory is particularly useful for designing distributed-element filters for use at microwave frequencies.

Microwave couplers are devices which divert a fraction of the signal on one transmission line to another transmission line.



https://www.microwaves101.com/encyclopedias/coupler-fundamentals

4.8. Aplicatii – modelarea microcomutatoarelor de RF

Microcomutatoarele MEMS de RF (RF MEMS switches) = dispozitive integrate ce folosesc o miscare mecanica pentru modificarea configuratiei unui circuit de RF

Exemplu de aplicatie

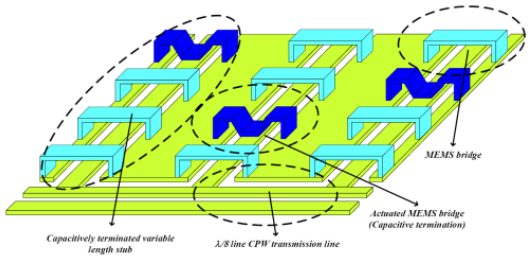
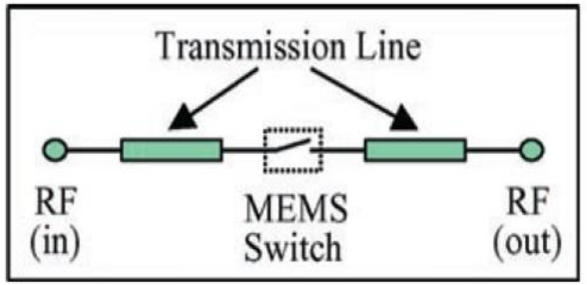


Figure 1.1 3D view of the MEMS variable impedance matching network.

[MehmetUnlu-PhD]

Plasare fata de TL:

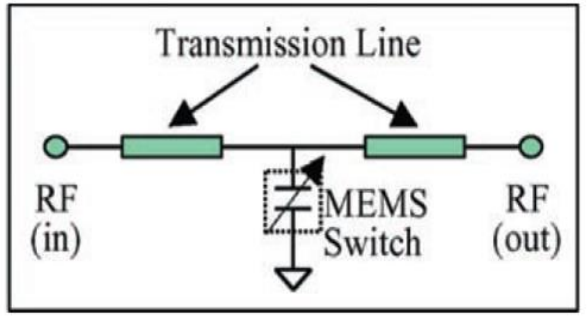
- Longitudinal (**serie ***)
- Transversal (**paralel ***)



(a)

Tip de contact

- Rezistiv (ohmic, galvanic)
- **Capacitiv***



(b)

Uzuale: (a) serie, rezistiv; (b) shunt, capacitiv
Dar se pot concepe: serie capacitiv si *shunt* rezistiv !

[Yao,2000]

4.8. Aplicatii – modelarea microcomutatoarelor de RF

Realizare

De tip ***punte=bara dublu incastrata (bridge)***

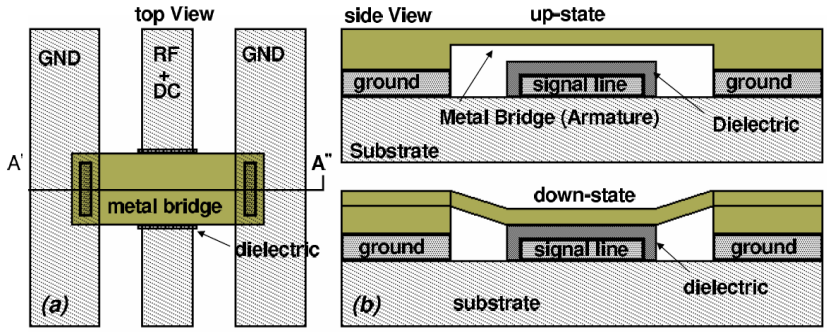


Figure 2.2: RF MEMS switch working principle. (a) Top view of the electrostatically actuated capacitive shunt device broad side implementation in CPW. (b) Device cross section along line $A' - A''$. Up - state allows the signal to pass through, while down state couples the input to ground.

Punte, *shunt*, capacitiv [Rangra-PhD]

De tip ***consola=bara simplu incastrata (cantilever)*** (bara incastrata la un capat, celalalt capat fiind liber)

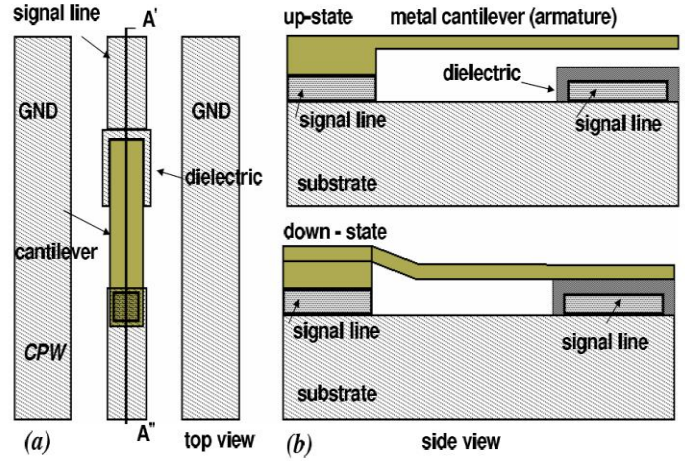


Figure 2.5: Electrostatically actuated capacitive series switch (inline) implemented in CPW configuration. (a) top view (b) side view showing the two states of the switch.

Consola, serie, capacitiv [Rangra-PhD]

4.8. Aplicatii – modelarea microcomutatoarelor de RF

Dispozitivele au doua stari stabile, care pot fi descrise din 3 puncte de vedere:

1. Al actionarii

- **Actionat** (activat de tensiunea de comanda)
- **Neactionat** (neactivat)

2. Al pozitiei membranei mobile fata de substrat (suprafata lui reprezinta un plan de referinta considerat orizontal)

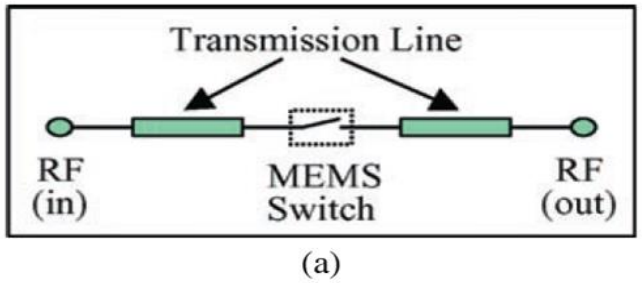
- **Sus**
- **Jos**

3. Al semnalului de RF care trece prin linie de transmisie (TL)

- **Semnalul** de RF **trece** prin TL
- **Semnalul** de RF **este blocat**

Asocierea dintre fiecare din cele 3 puncte de vedere depinde de tipul constructiv al *switch*-ului, si de modul de plasare al acestuia fata de linie (exemplu pe *slide*-ul urmator).

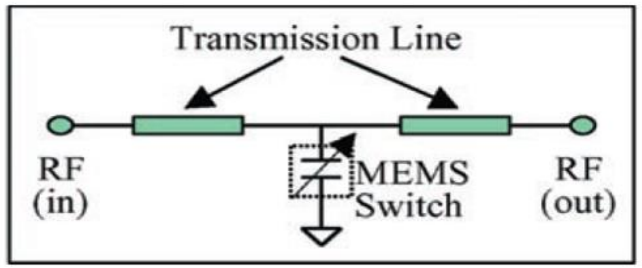
4.8. Aplicatii – modelarea microcomutatoarelor de RF



Serie

Stare stabila 1: actionat = jos = semnal RF trece

Stare stabila 2: neactionat = sus = semnal RF blocat



Shunt

Stare stabila 1: actionat = jos = semnal RF blocat

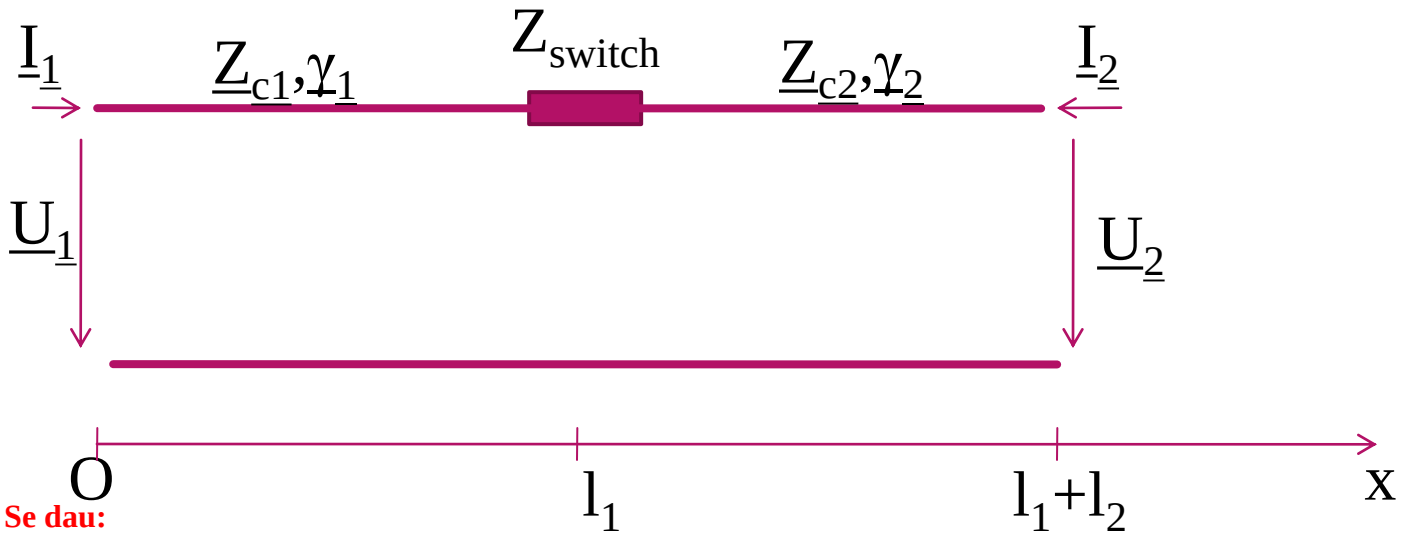
Stare stabila 2: neactionat = sus = semnal RF trece

[Yao,2000]

OBS 1: jos => distanta mica intre armaturi => capacitate mare => impedanta mica a switch-ului
(nu va deranja semnalul de RF la plasarea longitudinala; va face un scurt circuit la masa in plasarea transverala)

OBS 2: sus => distanta mare intre armaturi => capacitate mica => impedanta mare a switch-ului
(nu va deranja semnalul de RF la plasarea transverala; va reprezenta o intrerupere in plasarea longitudinala)

4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu



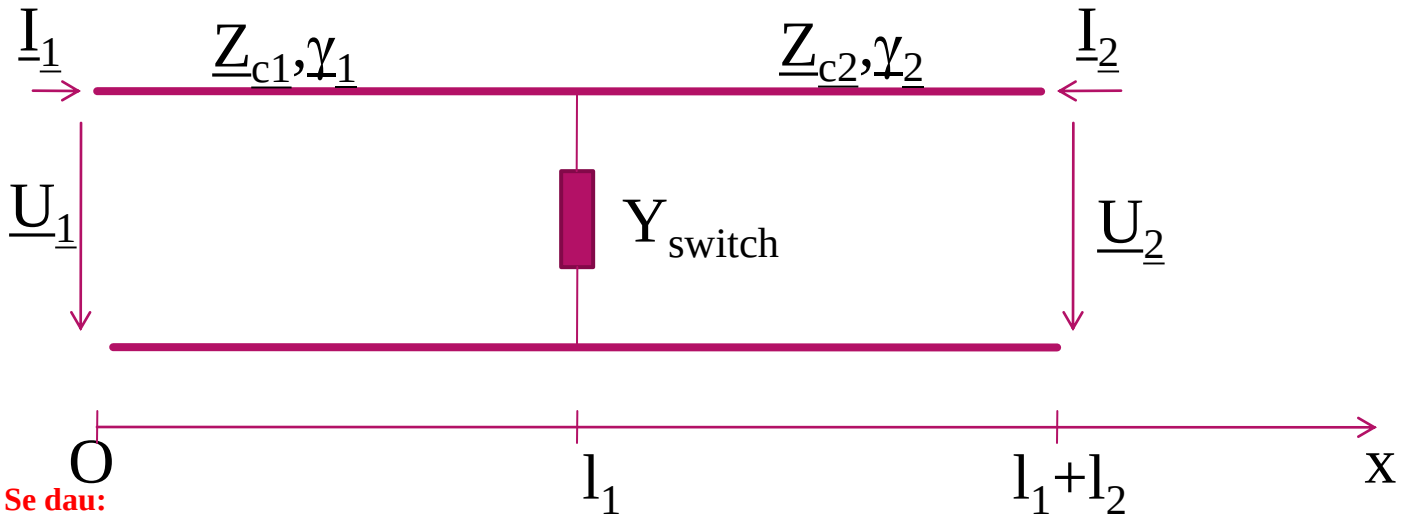
Se dau:

- parametrii secundari (sau parametrii lineici ai TLs)
- lungimea liniei
- gama de frecvente,
- impedanta (sau parametrii schemei echivalente cu parametri concentrati) switch-ului

Se cer:

- impedanta calculata fata de terminale
- tensiunea si curentul pe linie in conditiile in care se specifica ce se intampla la capete
- (ex: la capatul din stanga se alimenteaza cu o tensiune cunoscuta, la capatul din dreapta se pune o sarcina cunoscuta)

4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu



Se dau:

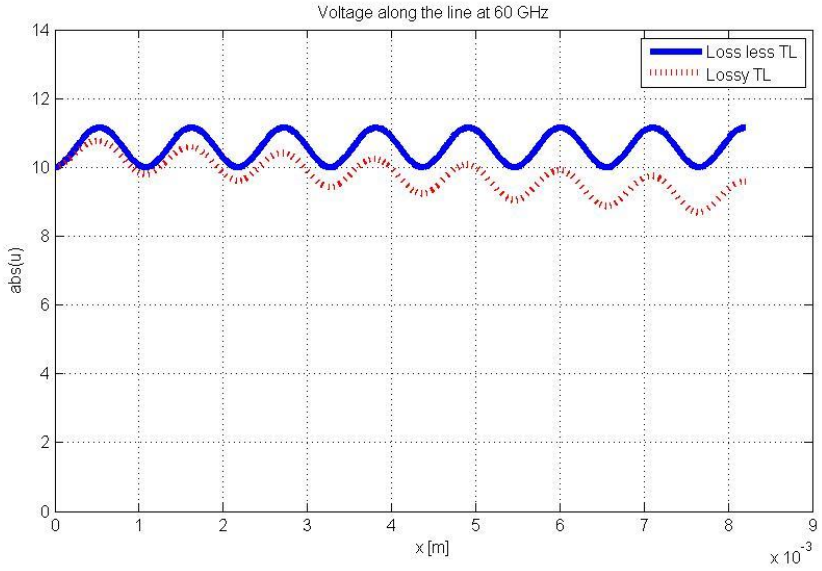
- parametrii secundari (sau parametrii lineici ai TLs)
- lungimea liniei
- gama de frecvente,
- impedanta (sau parametrii schemei echivalente cu parametri concentrati) switch-ului

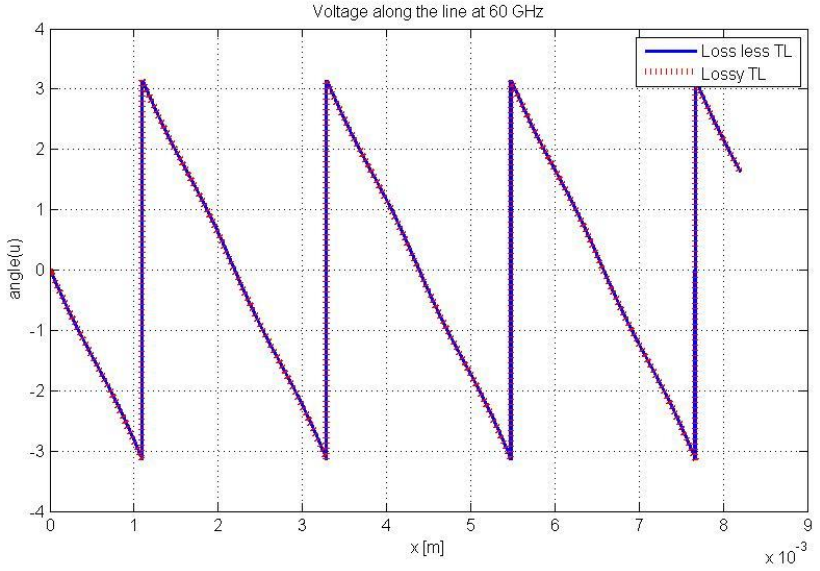
Se cer:

- impedanta calculata fata de terminale
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- (ex: la capatul din stanga se alimenteaza cu o tensiune cunoscuta, la capatul din dreapta se pune o sarcina cunoscuta)

Vom considera acest caz.

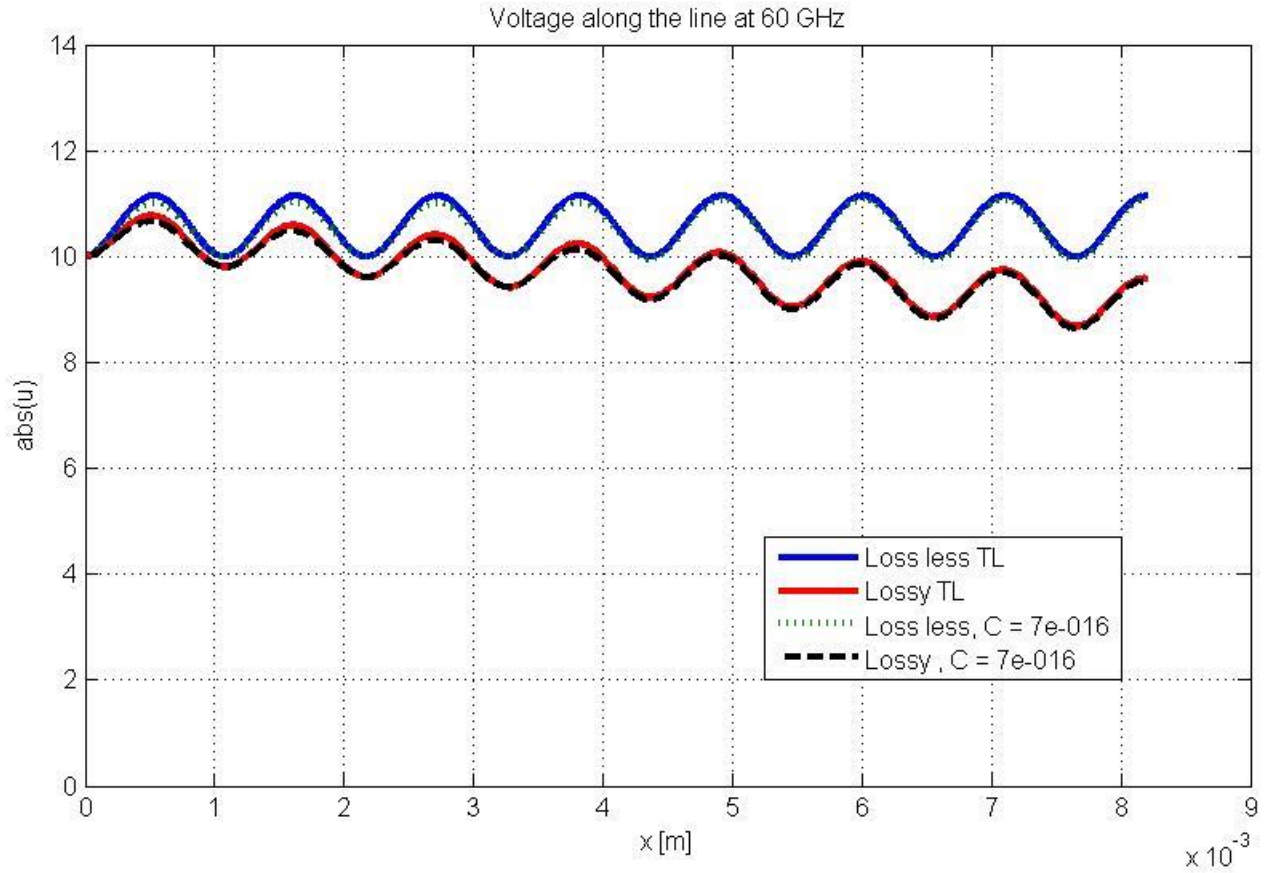
4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu

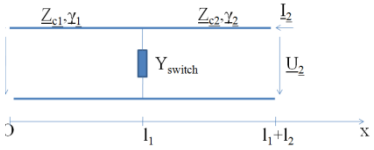




Tensiunea pe o linie electrica – cu si fara pierderi.

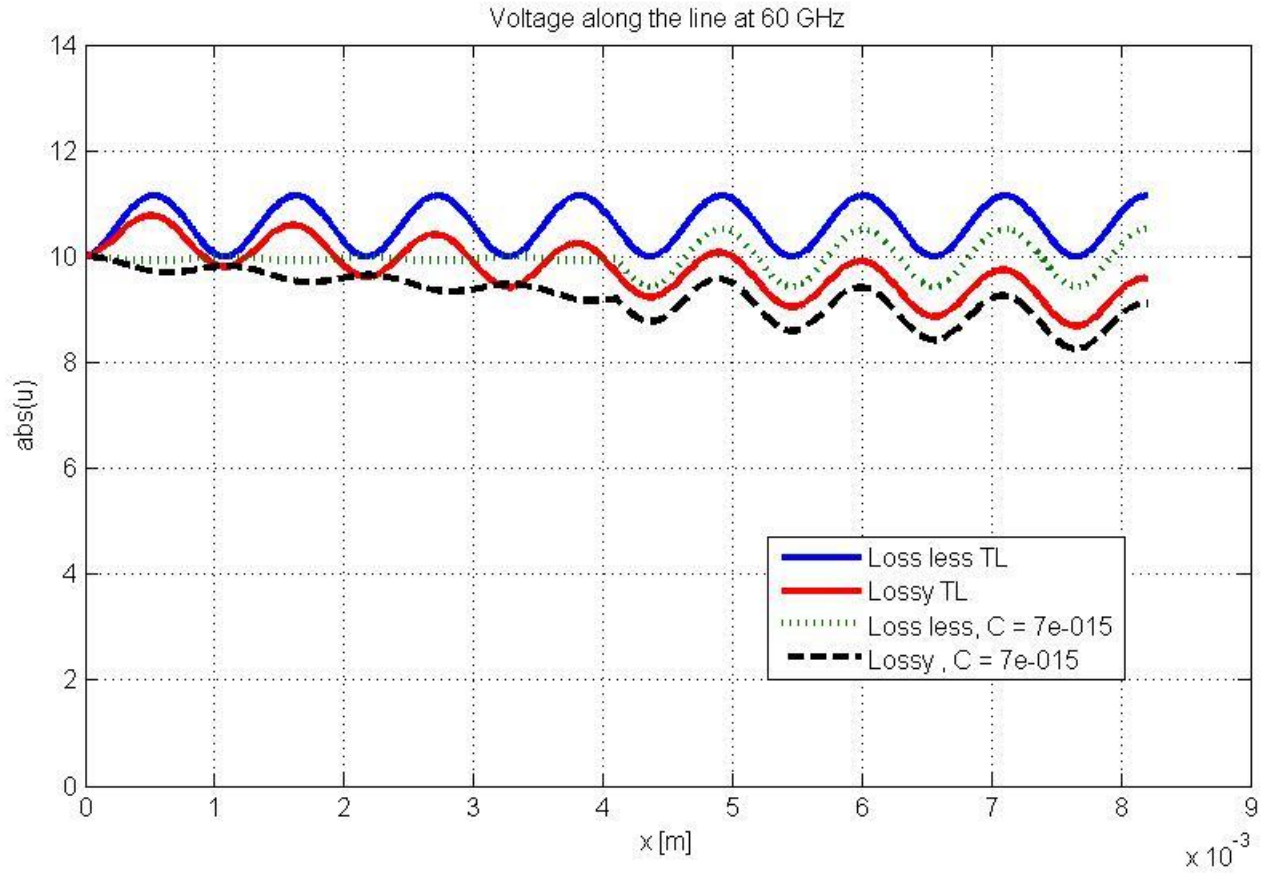
4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu

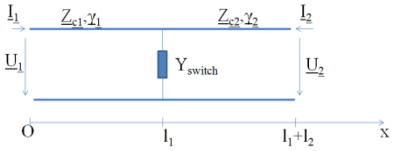




Tensiunea pe o linie electrica – un condensator introdus transversal la mijloc

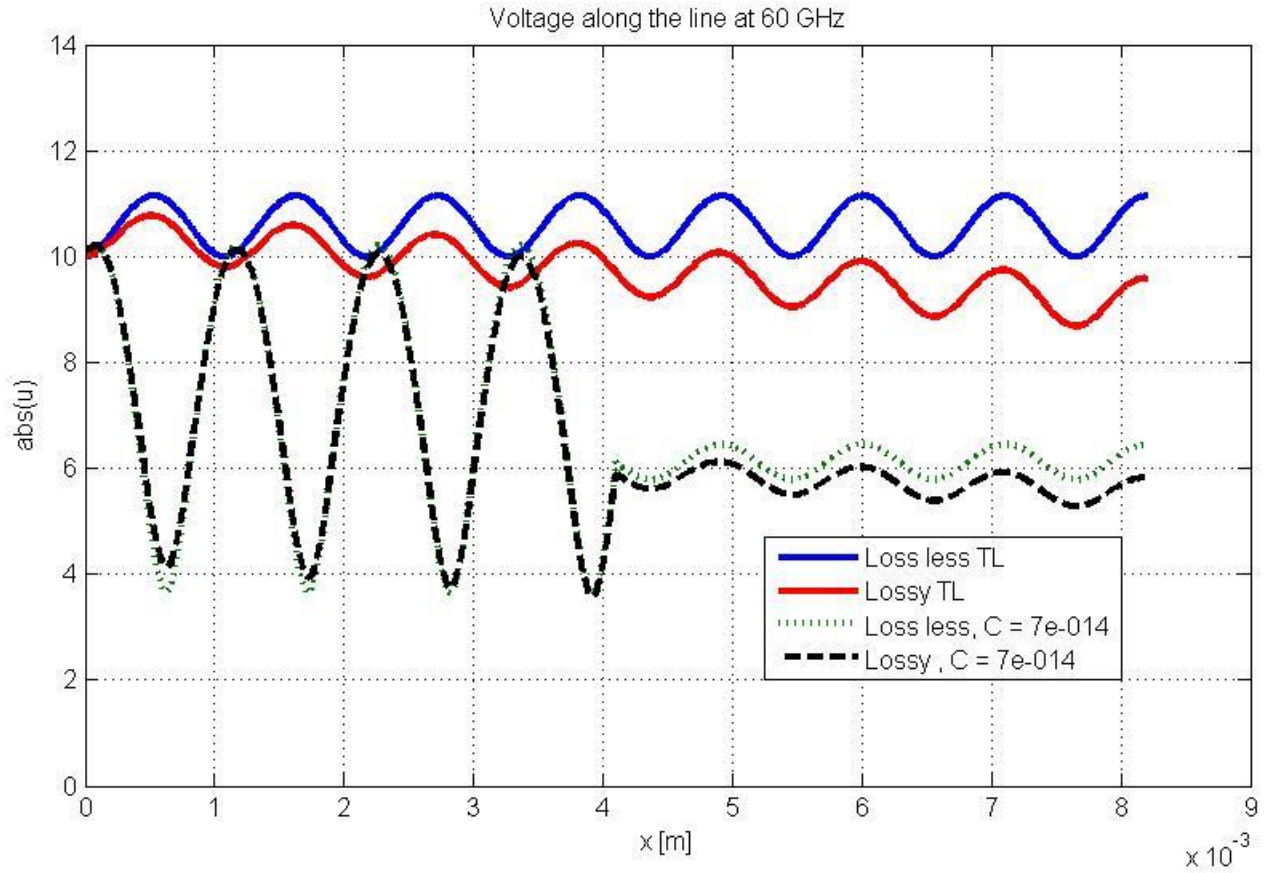
4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu

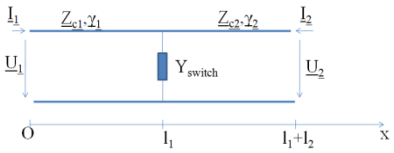




Tensiunea pe o linie electrica – un condensator introdus transversal la mijloc

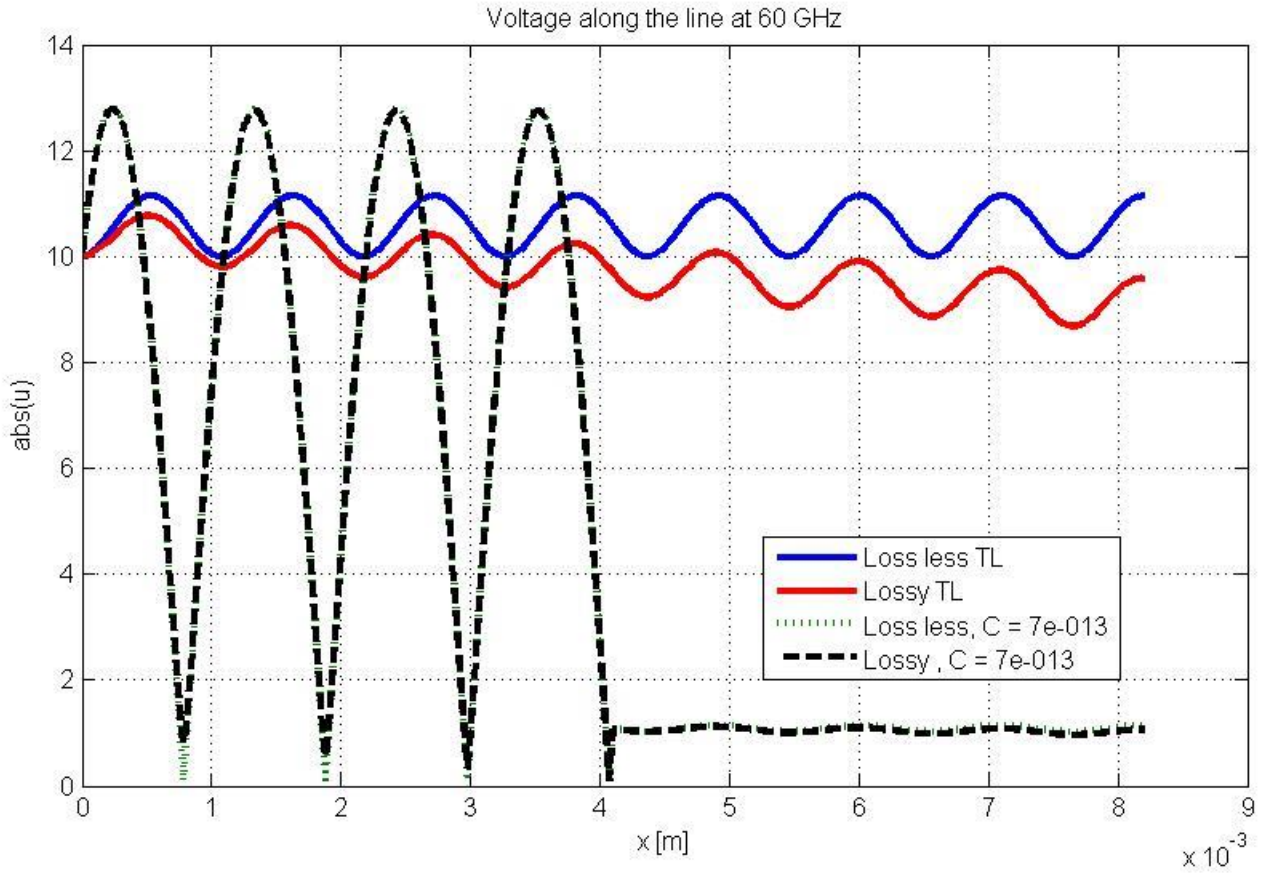
4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu

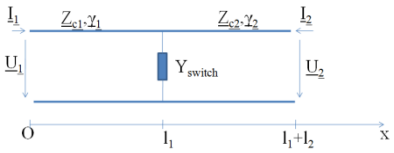




Tensiunea pe o linie electrica – un condensator introdus transversal la mijloc

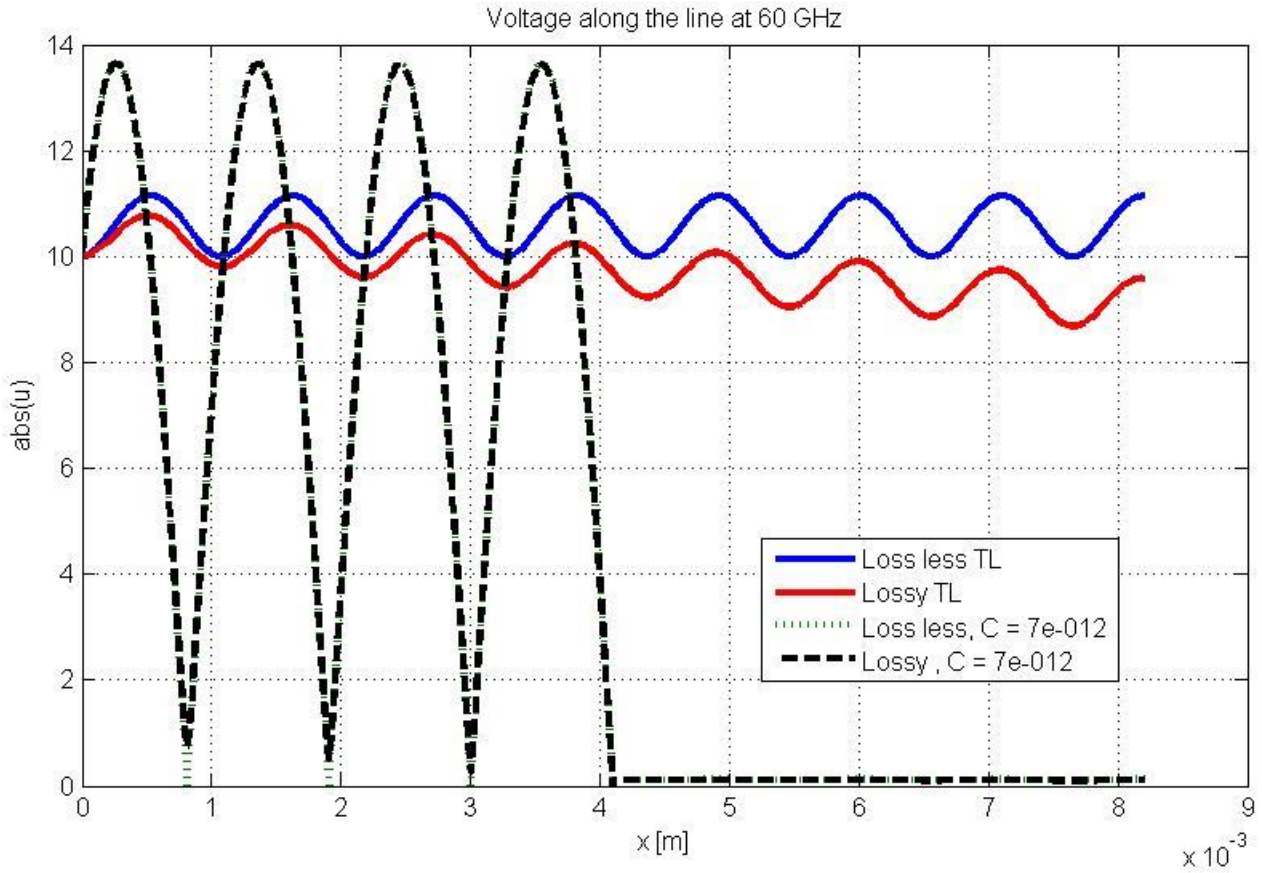
4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu

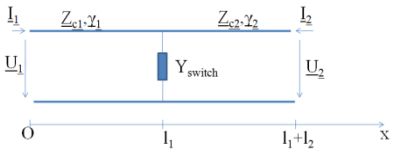




Tensiunea pe o linie electrica – un condensator introdus transversal la mijloc

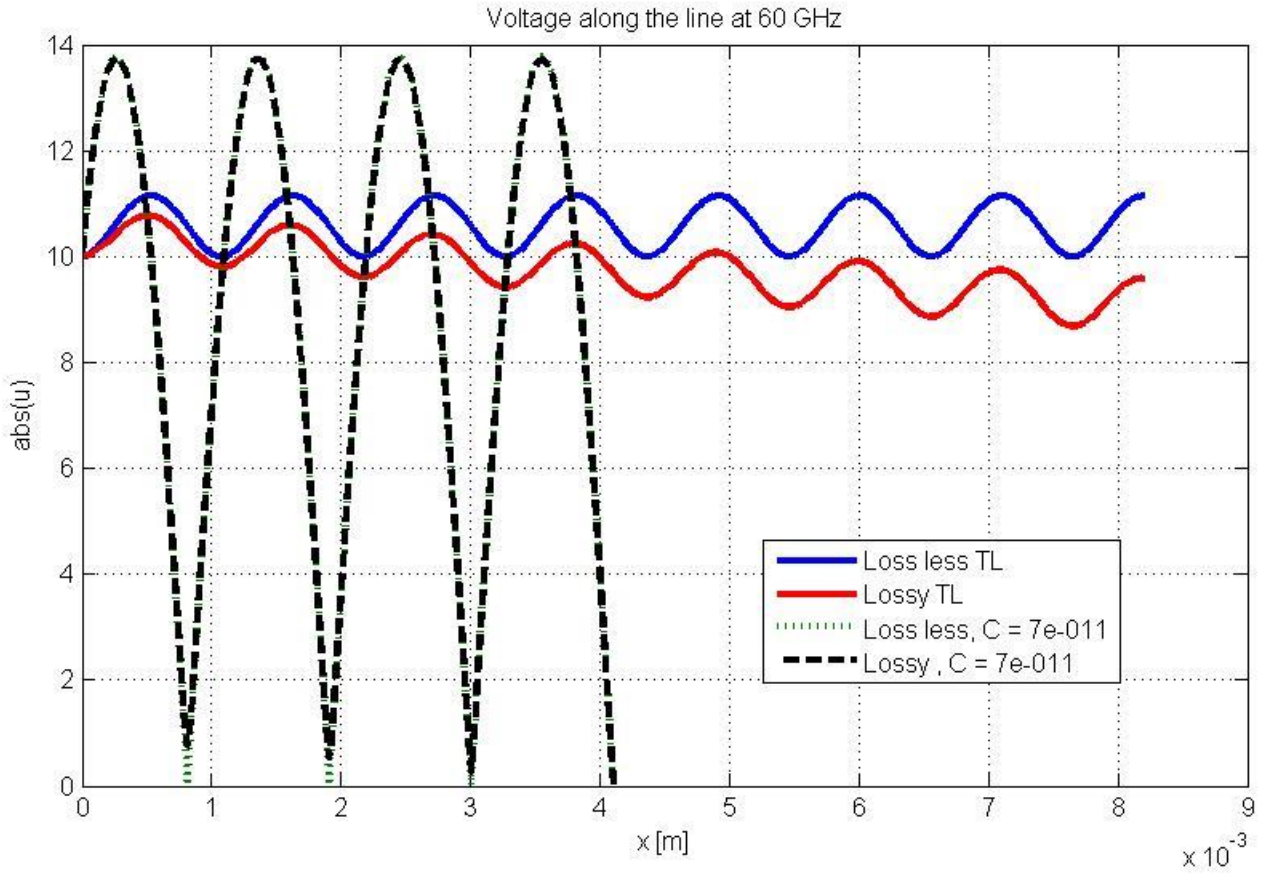
4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu

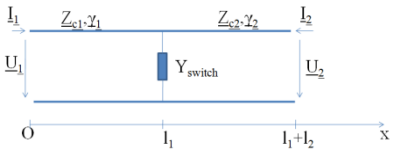




Tensiunea pe o linie electrica – un condensator introdus transversal la mijloc

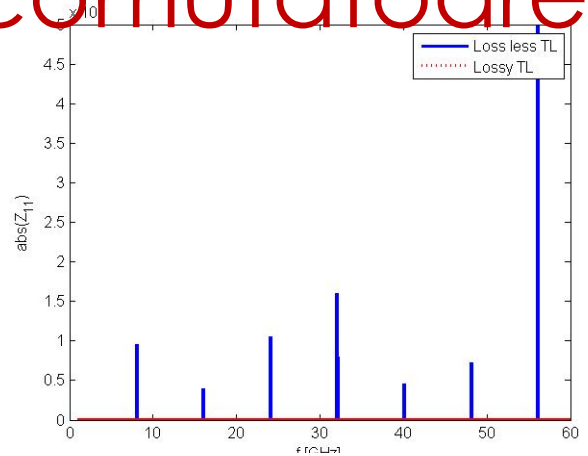
4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu

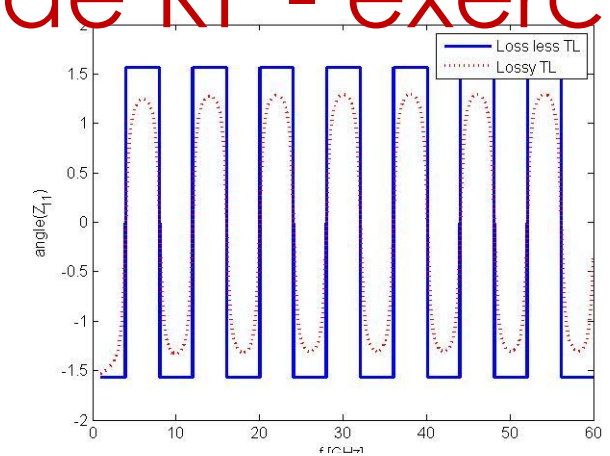


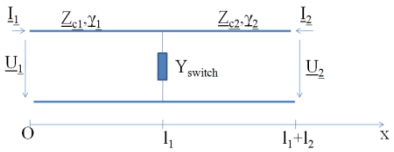


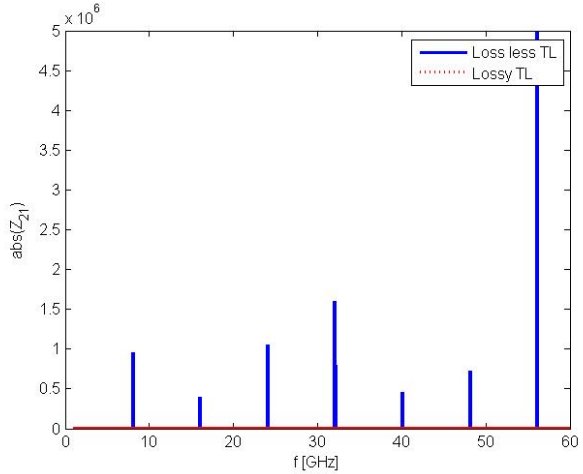
Tensiunea pe o linie electrica – un condensator introdus transversal la mijloc

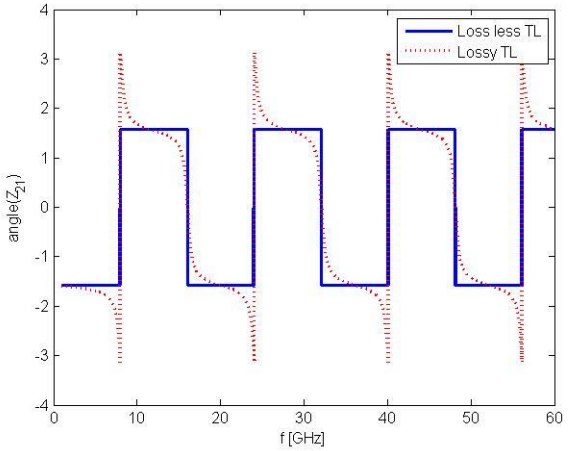
4.8. Aplicatii – modelarea microcomutatoarelor de RF - exercitiu







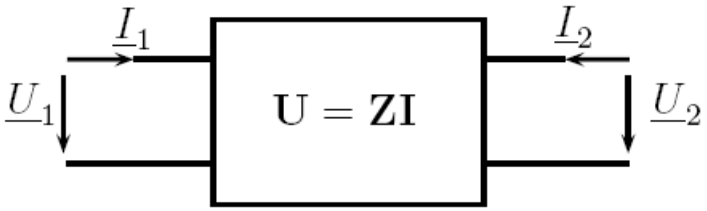


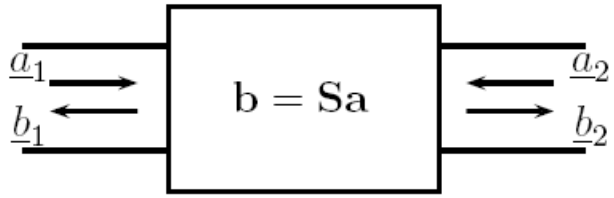


Caracteristica de frecventa – impedanta liniei, vazuta ca un sistem MIMO (2 intrari, 2 iesiri)

Aplicatii (facultativ) — Parametrii S

Parametrii S ("scattering", de repartitie, de imprastiere)





Marimile la borne sunt superpozitii de unde (indicele + pentru unda directa si – pentru unda inversa):

$$\begin{aligned} \underline{U}_k &= \underline{U}_{k+} + \underline{U}_{k-}, \\ \underline{I}_k &= \frac{1}{\underline{Z}_{0k}}(\underline{U}_{k+} - \underline{U}_{k-}). \end{aligned}$$

Rezulta ca:

$$\begin{aligned} \underline{U}_k &= \sqrt{\underline{Z}_{0k}}(\underline{a}_k + \underline{b}_k), \\ \underline{I}_k &= \frac{1}{\sqrt{\underline{Z}_{0k}}}(\underline{a}_k - \underline{b}_k). \end{aligned}$$

Se lucreaza cu valorile asociate:

$$\begin{aligned} \underline{a}_k &= \frac{\underline{U}_{k+}}{\sqrt{\underline{Z}_{0k}}}, \\ \underline{b}_k &= \frac{\underline{U}_{k-}}{\sqrt{\underline{Z}_{0k}}} \end{aligned}$$

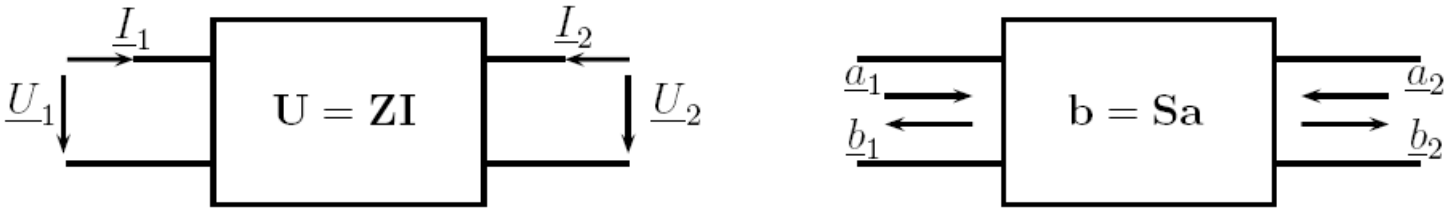
Puterea activa transferata prin poarta k

$$P_k = \mathrm{Re}\{\underline{U}_k \underline{I}_k^*\} = \mathrm{Re}\{\sqrt{\underline{Z}_{0k}}(\underline{a}_k + \underline{b}_k)\frac{1}{\sqrt{\underline{Z}_{0k}^*}}(\underline{a}_k^* - \underline{b}_k^*)\}.$$

In general Z_{ok} (impedanta caracteristica) este reala

Aplicatii (facultativ) — Parametrii S

Parametrii S



Daca impedanta caracteristica este pur reala atunci $P_k = a_k^2 - b_k^2,$

Prin definitie $\mathbf{b} = \mathbf{Sa},$

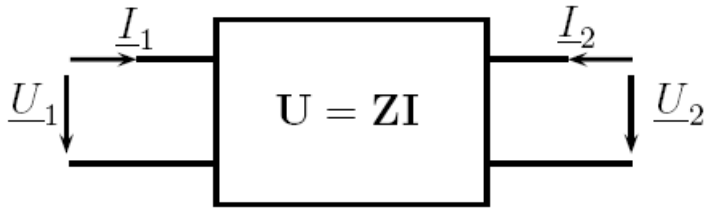
$$\underline{S}_{ii} = \left.\frac{\underline{b}_i}{\underline{a}_i}\right|_{\underline{a}_j=0, j\neq i}$$ Coeficient de reflexie la portul i

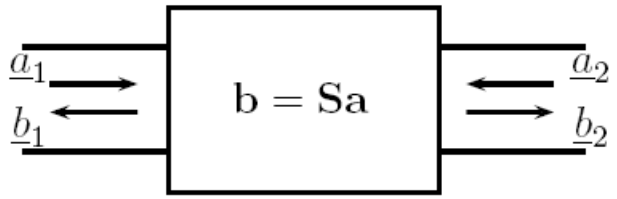
$$\underline{S}_{ij} = \left.\frac{\underline{b}_i}{\underline{a}_j}\right|_{\underline{a}_k=0, k\neq i}$$ Coeficient de transmisie de la portul j la portul i

ak si bk descriu puterile undei diracta si inversa (se masoara in $\sqrt{W}$).

Aplicatii (facultativ) — Parametrii S

Parametrii S





$$|\underline{S}_{ii}|^2 = \left| \frac{\underline{b}_i}{\underline{a}_i} \Big|_{\underline{a}_j=0, j\neq i} \right|^2$$

Coeficient de reflexie al puterii

$$|\underline{S}_{ij}|^2 = \left| \frac{\underline{b}_i}{\underline{a}_j} \Big|_{\underline{a}_k=0, k\neq i} \right|^2$$

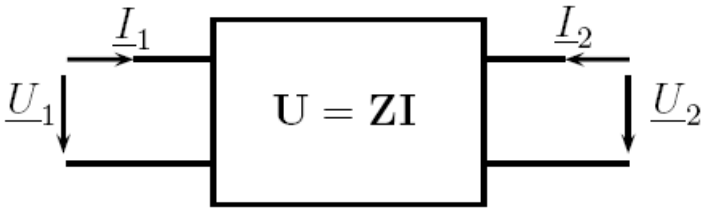
Coeficient de transmisie al puterii (randament) de la portul j la portul i

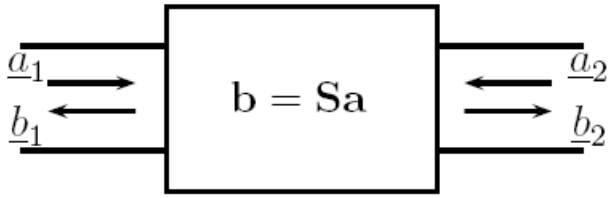
Conversii S-Z

$$\mathbf{S} = (\mathbf{Z} - Z_0\mathbf{1})(\mathbf{Z} + Z_0\mathbf{1})^{-1}, \qquad \mathbf{Z} = (\mathbf{1} - \mathbf{S})^{-1}(\mathbf{1} + \mathbf{S})Z_0,$$

Aplicatii (facultativ) — Parametrii S

Parametrii S - Proprietati





Dispozitive liniare si reciproce – S este simetrica.

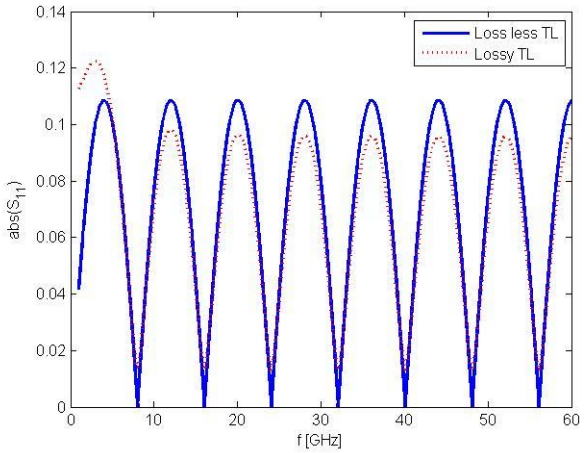
Dispozitive pasive $\mathbf{I} - \mathbf{S}^H\mathbf{S}$ este pozitiv definita pentru orice frecventa

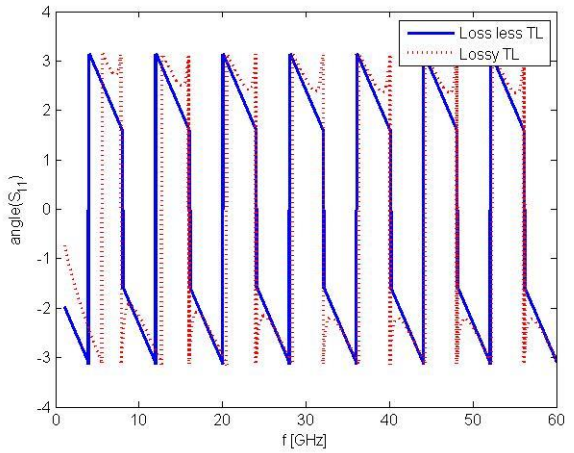
rezulta $|S_{ji}| \leq 1 \ \forall i, j.$

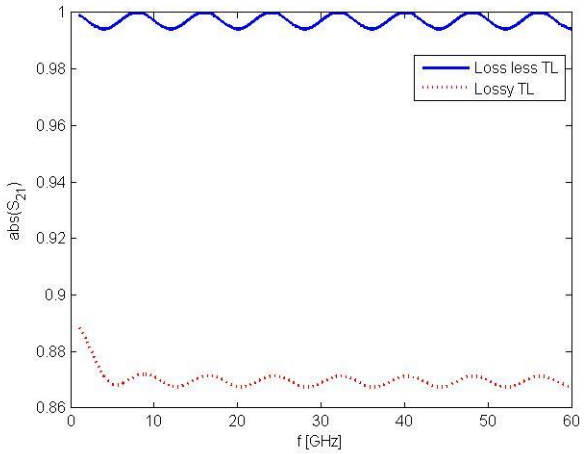
Dispozitive pasive fara pierderi – S este ui $\mathbf{I} - \mathbf{S}^H\mathbf{S}$

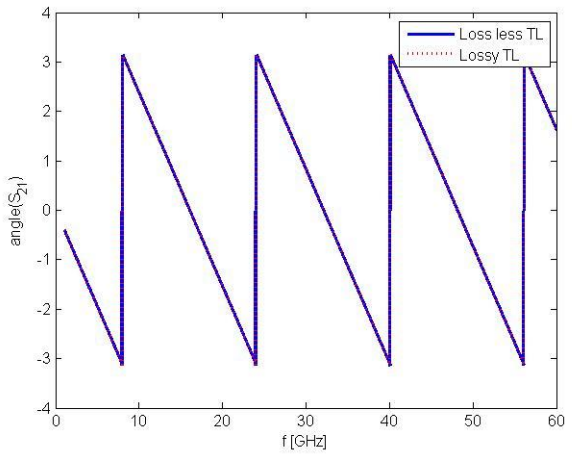
Aplicatii (facultativ) — Parametrii *S*

Parametrii S – pentru o linie simpla



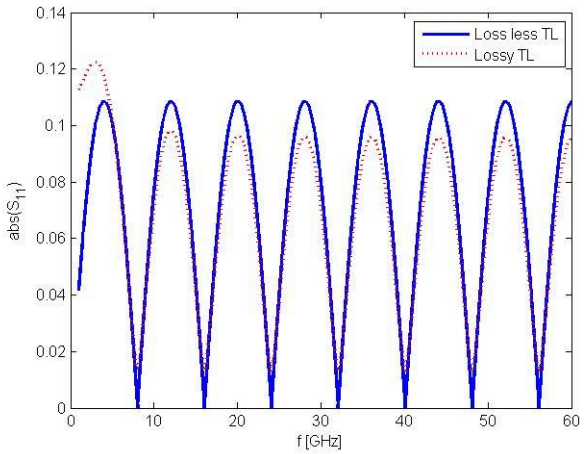


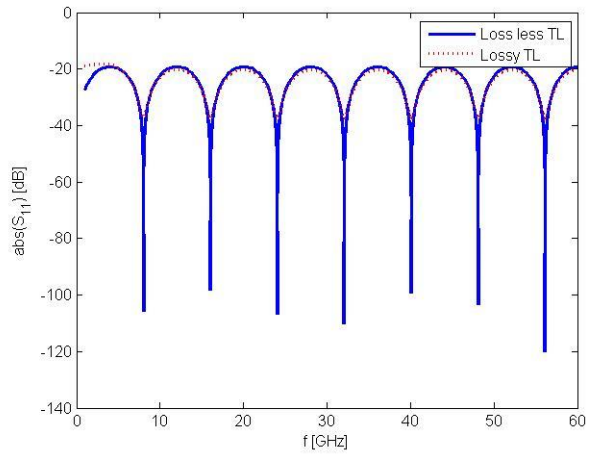


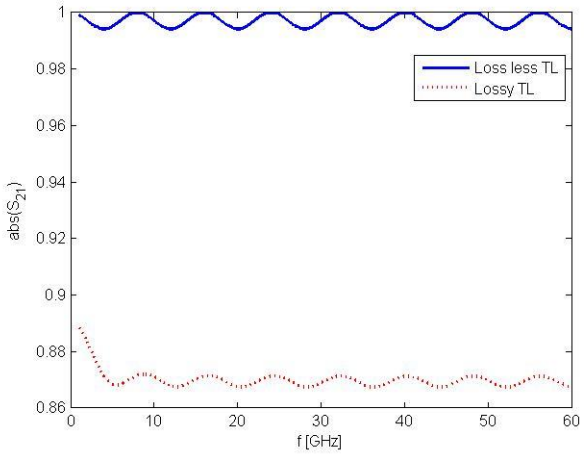


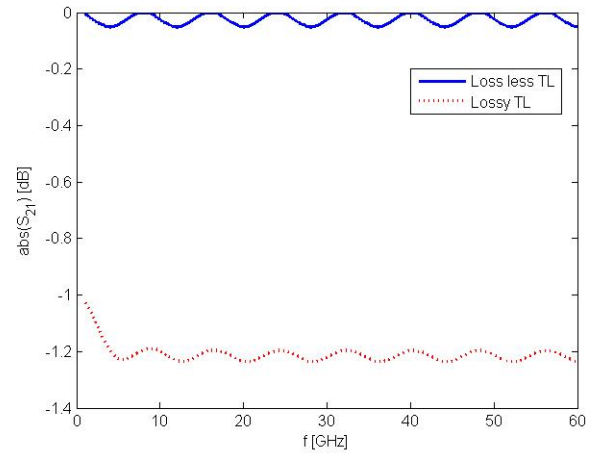
Aplicatii (facultativ) — Parametrii S

Parametrii S – pentru o linie simpla – modulul in dB



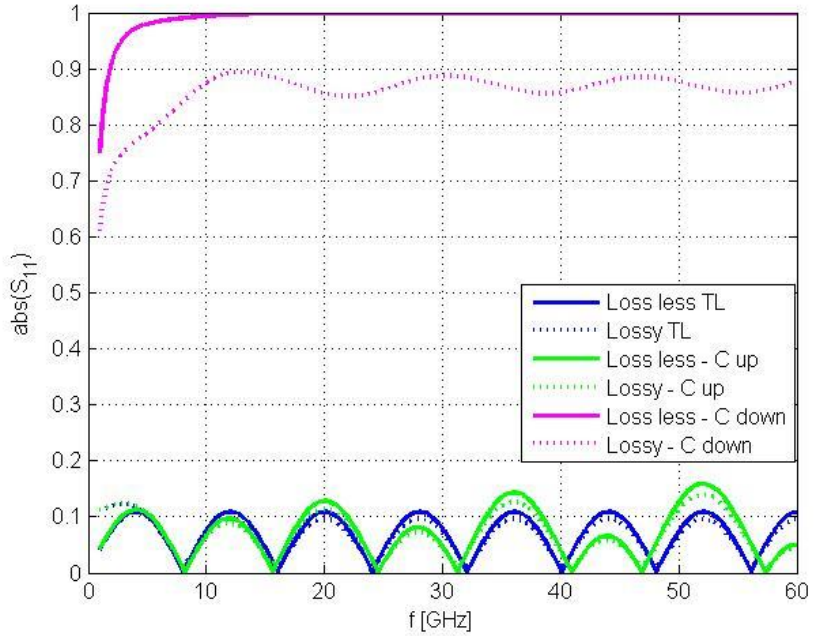


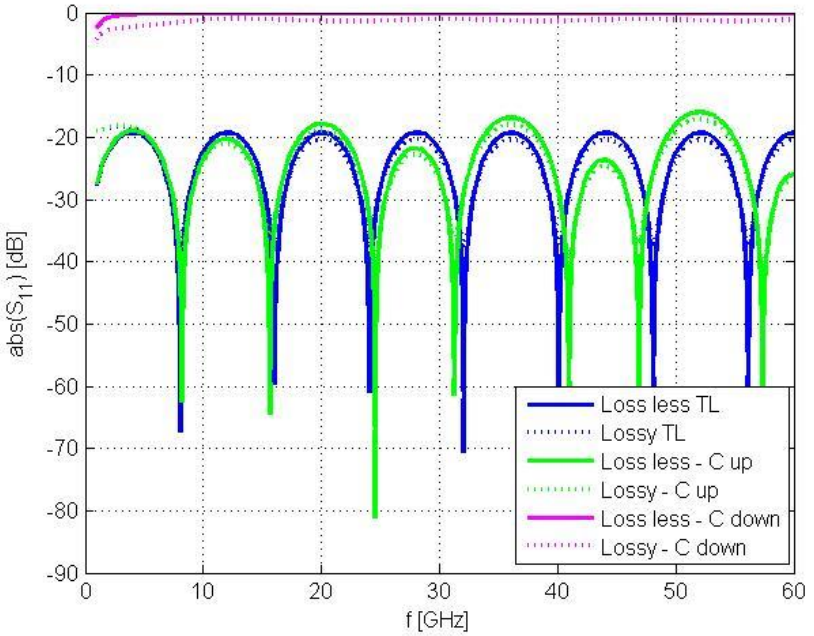


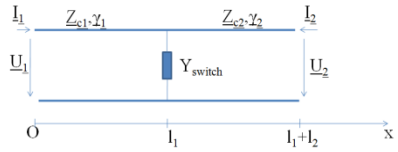


Aplicatii (facultativ) — Parametrii S

Parametrii S – efectul introducerii unui Cup = 7e-15 (on) si Cdown (off) = 7e-12

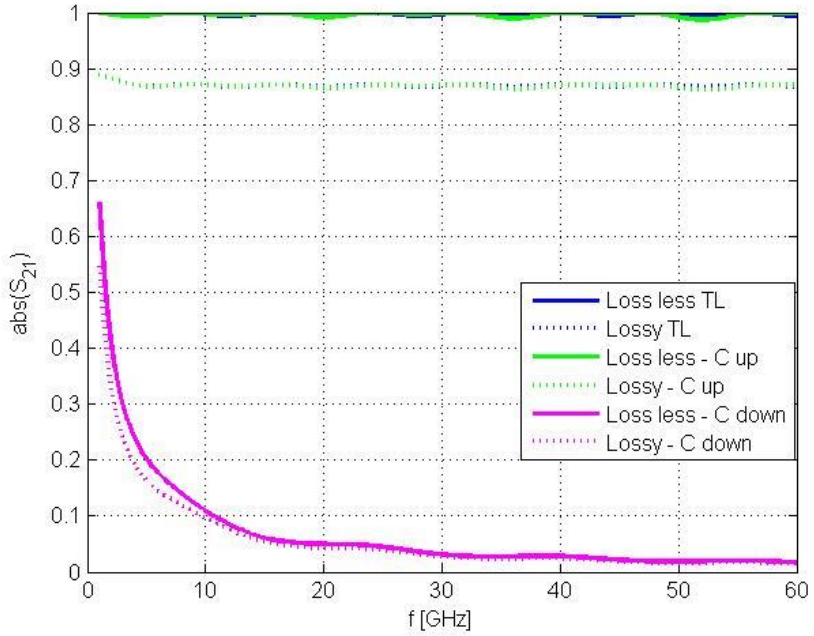


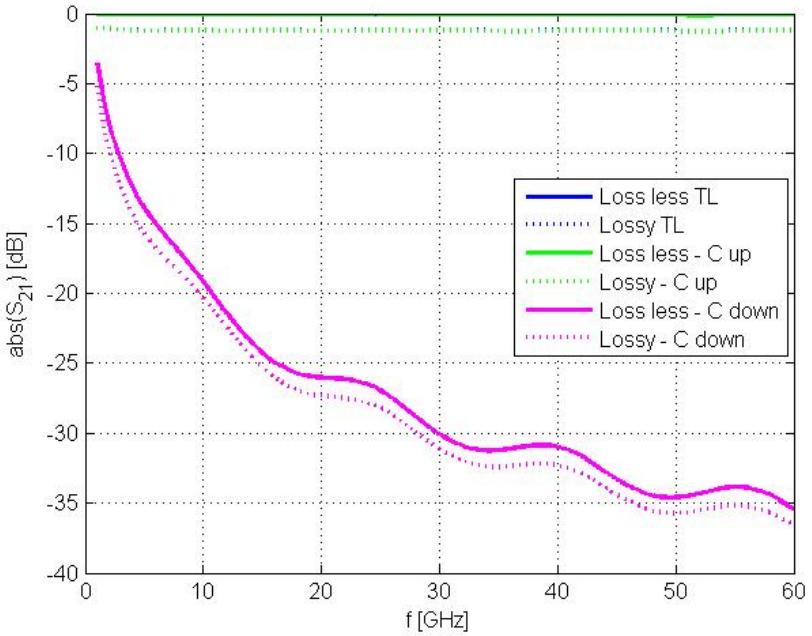


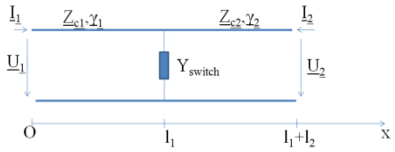


Aplicatii (facultativ) — Parametrii S

Parametrii S – efectul introducerii unui Cup = 7e-15 (on) si Cdown = 7e-12 (off)







Aplicatii (facultativ) — Parametrii S

Performantele unui switch dpdv al semnalului de RF

1. Pierderi prin insertie = pierderi rezitive pe linie, incluzand efectul pelicular, in starea in care semnalul de RF trece prin linie
= S21 in starea *pass-through*

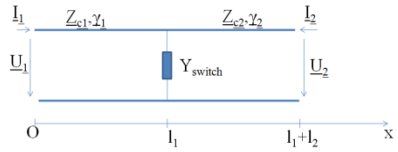
2. Pierderi prin reflexie = pierderi datorate reflexiei semnalului transmis, datorate neadaptarii sarcinii in starea in care semnalul de RF trece prin linie
= S11 in starea *pass-through*

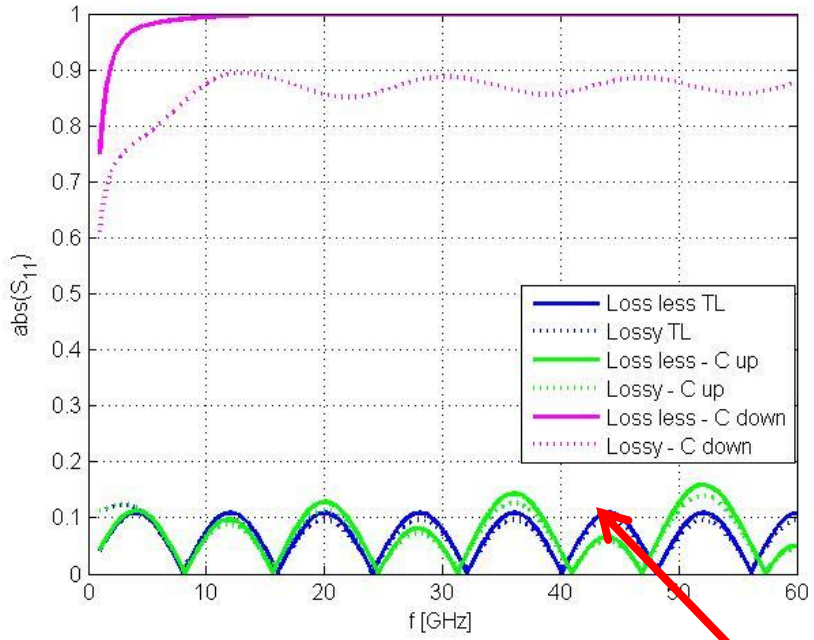
3. Izolarea = izolarea intre portul de intrare si cel de iesire, in starea in care semnalul de RF este blocat
= S21 in starea *blocking*

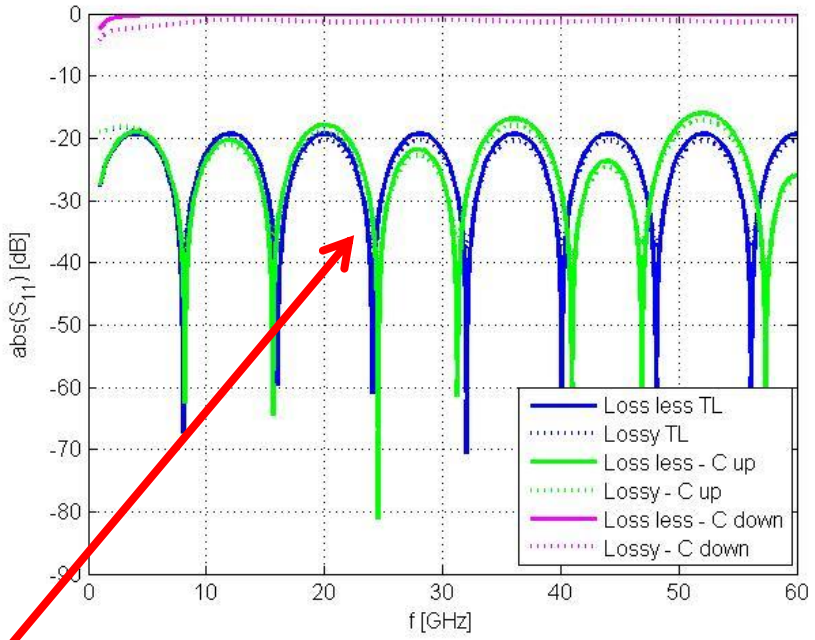
Dictionar:
Pierderi prin insertie = *Insertion losses*
Pierderi prin reflexie = *Return losses*
Izolare = *Isolation*

Aplicatii (facultativ) — Parametrii S

Parametrii S – efectul introducerii unui Cup = 7e-15 (on) si Cdown (off) = 7e-12





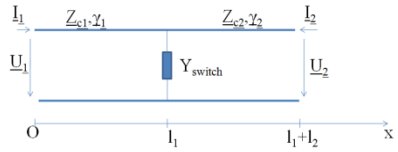


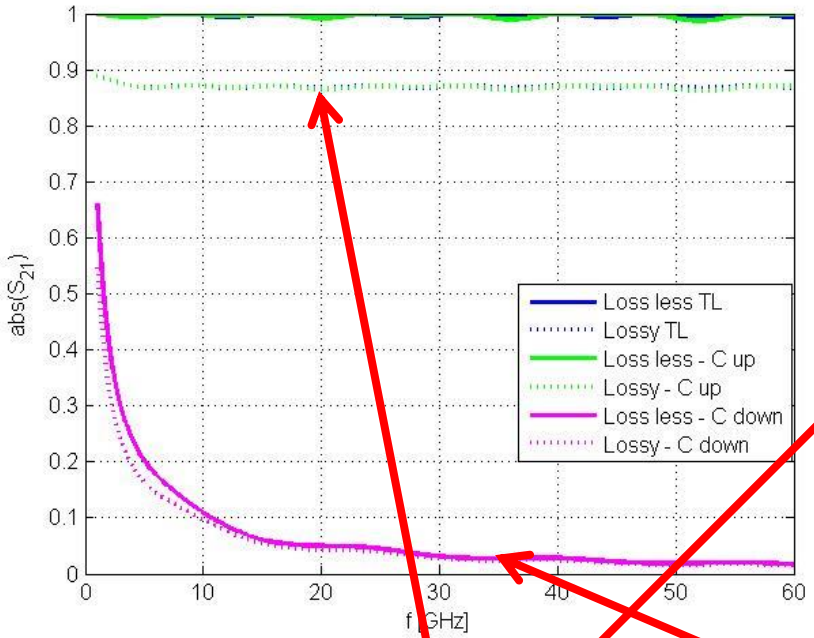
Return losses

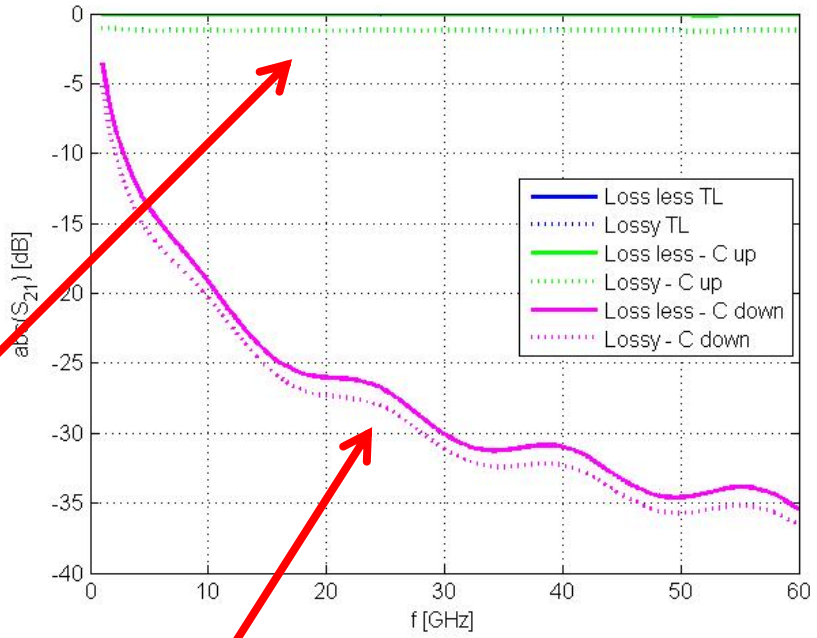
Condensator plasat transversal fata de linie

Aplicatii (facultativ) — Parametrii S

Parametrii S – efectul introducerii unui Cup = 7e-15 (on) si Cdown = 7e-12 (off)





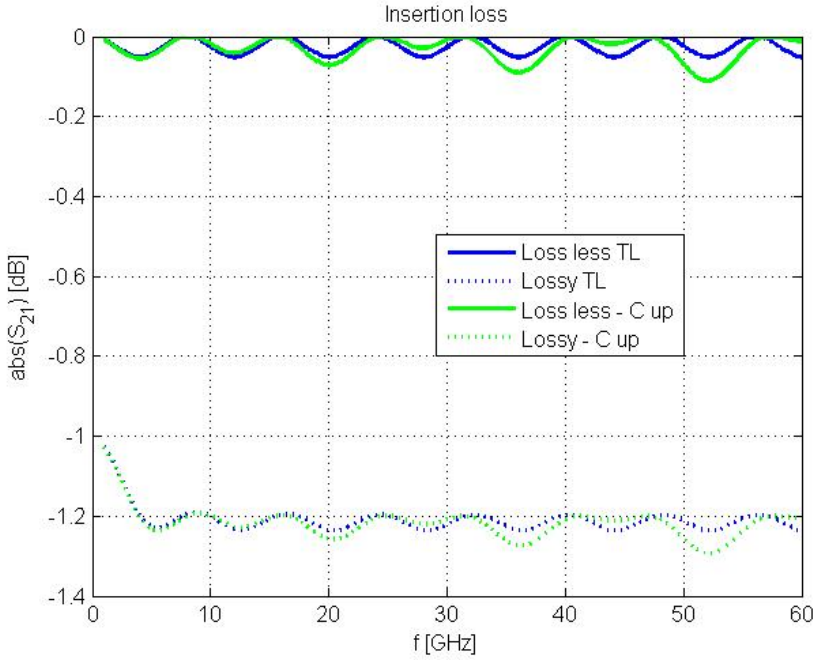


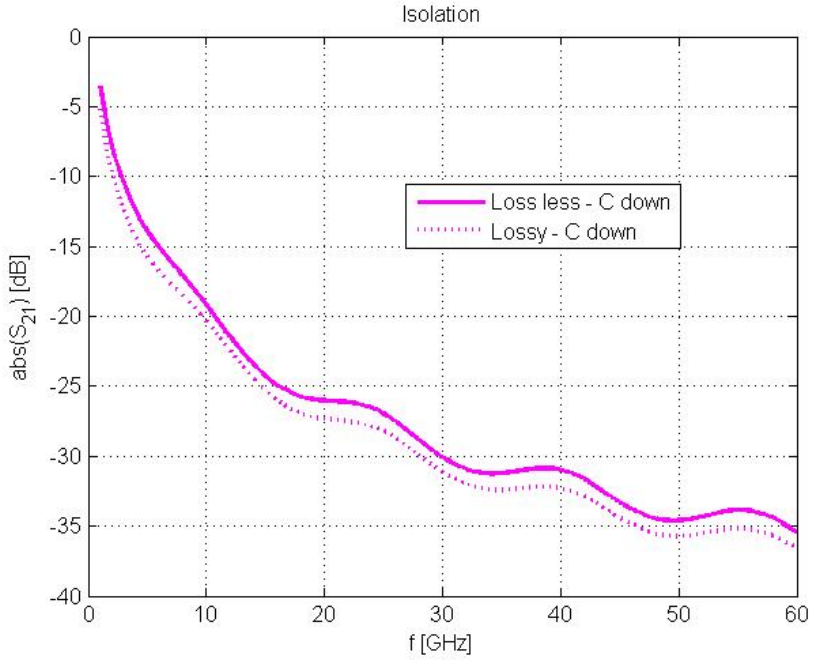
Insertion losses **Isolation**

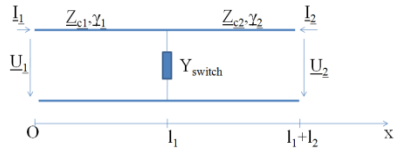
Pierderile prin insertie si izolarea NU SE PUN IMPREUNA PE ACELASI GRAFIC!

Aplicatii (facultativ) — Parametrii S

Parametrii S – efectul introducerii unui Cup = 7e-15 (on) si Cdown = 7e-12 (off)







Insertion losses

Isolation

(ordine de marime – ca cele raportate prin lucrari)

Aplicatii (facultativ) — Parametrii S

Parametrii S – se refera la *switch*-ul conectat la o portiune scurta de linie

Valori tipice pentru un switch capacitiv de tip shunt
• Insertion losses: -0.15 dB la10 GHz, -0.28 dB la 35 GHz,
• Isolation: −15 dB at 10 GHz and −5 dB at 35 GHz

Atentie, uneori prin lucrari nu se pune semnul minus, desi toate valorile lui S sunt < 1 deci in dB sunt negative.

Se doreste
• "low insertion losses" (in modul cat mai apropiate de 1, deci in dB cat mai aproape de 0 dB).
• "high isolation" (in modul cat mai aproape de 0, deci in dB cat mai mare in valoare absoluta).

Un switch ideal are 0 dB insertion loss si infinite isolation. ("low" si "high" se refera la valorile in dB, luate cu semnul +)

Parametrii switch-ului din schema de circuit (cu parametri concentrate si distribuiti) se determina din modelari electromagnetice.

Aplicatii (facultativ) — Parametrii S

Parametrii S sunt importanti pentru ca ei sunt cei care se masoara.

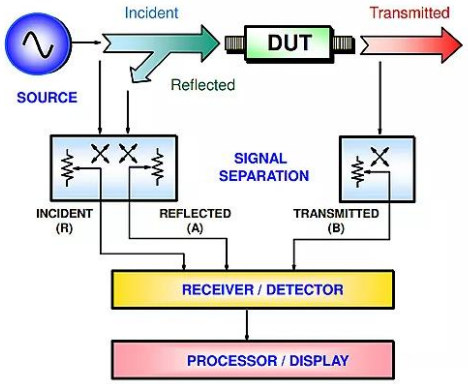
https://www.keysight.com/us/en/solutions/measurement-fundamentals/network-analysis.html#WhatIs

Vector network analyzers (VNAs) are test instruments that measure electrical network parameters. They are essential for radio frequency (RF) and microwave component analysis of various passive and active devices including filters, antennas, and power amplifiers.



Today, most network analyzers are vector network analyzers — measuring both magnitude and phase. Vector network analyzers are extremely versatile instruments that can characterize S-parameters, match complex impedances, make time-domain measurements, and more.

The network analyzer's receivers measure the incident, reflected, and transmitted signals to calculate the forward S-parameters.



General Network Analyzer block diagram

Concluzii

- Liniile de transmisie (cabluri, interconexiuni) sunt folosite pentru transmiterea ghidata a semnalelor de la surse la sarcini (Conf.T.energiei-Poynting, puterea se transmite prin dielectric, conductoarele doar ghideaza).
- Daca linia are o lungime comparabila cu lungimea de unda corespunzatoare frecventei de lucru, semnalele nu se transmit instantaneu, ele se propaga cu intarzieri si perturbatii.

- In acest capitol am studiat liniile simple, care au doua conductoare. Ele pot fi private ca elemente cuadripolare de dip diport, liniare, pasive, reciproce, de circuit electric cu parametri distribuiti. Matricile de impedanta si admitanta sunt simetrice (elementul este reciproc). Daca liniile sunt omogene (au aceleasi dimensiuni si materiale de-a lungul lor) atunci $\underline{Z}_{11} = \underline{Z}_{22}$, $\underline{Y}_{11} = \underline{Y}_{22}$. La un element cuadripolar de tip diport, alcatuit numai din elemente de circuit cu parametri concentrati, componentele impedantelor, de exemplu $\underline{Z}_{11}$sunt functii rationale cu un numar de poli egal cu numarul elementelor reactive (bobine si condensatoare). La linia de transmisie, $\underline{Z}_{11}$are o infinitate de poli si zerouri, care alcatuiesc un sir divergent (cu punct de acumulare la infinit). Intr-o banda finite de frecvente $(f_{\min}, f_{\max})$ se afla un numar finit de frecvente de rezonanta, dependent de raportul $\frac{l}{\lambda}$. Cu cat $f_{\max}$ e mai mare, cu atat sunt mai multi poli in acest interval, deci linia se poate echivala cu un numar mai mare de tronsoane LC. De regula, fiecare lungime de unda se discretizeaza in 10 tronsoane de tip Γ, cu $L = \frac{L_l \lambda}{10}$ longitudinal si $C = \frac{C_l \lambda}{10}$ transversal, dar aceasta modelare nu este cea mai buna aproximare posibila.

- Aplicatie principala discutat – **adaptarea** sarciniilor la linii (“**impedance matching**”), fara de care semnalele ajung la sarcini perturbate, aparand reflexii nedorite. In plus, linia adaptata este cea mai eficienta, transmite puterea maxima. Parametrii lineici ai liniilor se obtin din datele geometrice si de material, cu rationamente care se bazeaza pe capitolele 1,2,3. Ei sunt cei responsabili pentru **intarzierea si perturbarea semnalului** transmis catre sarcini, si **limiteaza frecventa maxima** la care un cablu poate fi folosit pentru transmiterea semnalelor digitale.
- O alta aplicatie importanta a liniilor este cea a folosirii lor ca **elemente ideale de circuit LC in circuitele cu microunde**. Din motive de timp, acest subiect nu a putut fi tratat in detaliu in aceasta prezentare.

Referinte

A. Tomescu si F.M.G Tomescu, Sisteme cu Microunde, Editura MatrixROM (o pot imprumuta celor interesati)

A. Timotin, Lectii de bazele electrotehnicii, 1970, cap 53 si 54 (disponibila pe pagina moodle)

M. Sadiku, Elements of electromagnetics, 3rd edition, cap 11 (disponibila pe pagina moodle)

Sophocles J. Orfanidis, Electromagnetic Waves and Antennas, 2016 disponibila online la https://www.ece.rutgers.edu/~orfanidi/ewa/ (cap 11 si 16)

Steven W. Ellingson – Electromagnetics (I), 3. Transmission Lines – online la
https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Electro-Optics/Book%3A_Electromagnetics_I_(Ellingson)/03%3A_Transmission_Lines

Jin Au Kong – Electromagnetic Wave Theory – EMW Publishing, disponibila la
https://www.piers.org/piers/files/BOOK_ElectromagneticWaveTheory_by_JINAUKONG_FreeRelease2021.pdf